Implicit Doubly Stochastic Graph-Based Subspace Clustering for Hyperspectral Band Selection

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***Abstract*—Band selection plays a critical role in reducing data redundancy of hyperspectral images (HSIs), where subspace clustering-based methods have shown remarkable potential, owing to the effective extraction of low-dimensional representations. Despite conducting band grouping competently, they not only overlook preserving original affinity structures but also require a manual graph normalization step, which could lead to suboptimal performance. To solve these issues, we propose a novel band selection method termed implicit doubly stochastic graph-based subspace clustering (IDSGC), facilitating end-to-end band affinity graph construction. First, a latent bipartite graph modeling (LBGM) module is designed to decompose the inherent band similarities into a bipartite graph, while mitigating the interference of noise links. Second, a novel implicit learning mechanism (ILM) of a doubly stochastic graph is presented to refine graph structure and low-dimensional representations simultaneously, which seamlessly integrates the LBGM with subspace learning. Finally, a probability-constrained affinity graph that is implicitly normalized during optimization is generated, which can be directly used for band clustering without additional postprocessing. Extensive experiments demonstrate that IDSGC outperforms several state-of-the-art band selection methods.**

***Index Terms*—Doubly stochastic graph, graph factorization, hyperspectral band selection, subspace clustering.**

I. Introduction

HYPERSPECTRAL imaging technology captures detailed spectral signatures of ground targets [1], providing massive hyperspectral image (HSI) data for applications such as Earth observation, mineral exploration, and agriculture [2], [3], [4]. HSI records a specific area with rich spatial and spectral information across numerous contiguous narrow bands [5], which may lead to the curse of dimensionality problem. The disparity between the data volume required for reliable model estimation and the limited number of available training pixels [6], i.e., the Huges phenomenon, often restricts the efficiency of downstream HSI analysis tasks. Hence, the dimensionality reduction is intensively researched, mainly consisting of the feature extraction and band selection strategies [7]. The former reconstructs several features in a low-dimensional space, while the latter identifies and retains a subset of the original bands [8]. The band selection approach maintains the physical meaning of the spectral dimensions in the original HSI [9], making it well-suited for HSI dimensionality reduction.

Depending on the availability of labeled data, band selection methods can be categorized into supervised [10], [11], and unsupervised ones [12], [13]. As annotating HSI pixels is significantly laborious and expensive [14], unsupervised methods are more appreciated in the unlabeled scenario, where the evaluation of individual band utility is the focus [15]. Different evaluation strategies give rise to different band selection methods [16], including searching-based, ranking-based, and subspace-learning-based approaches. Regarding the searching-based methods [17], [18] iteratively recombine a candidate band subset to satisfy a predefined evaluation function. However, the combinatorial explosion of subsets results in high computational complexity and suboptimal solutions [19]. As for the ranking-based band selection methods, a score is assigned to each band according to specific metrics, and the top-ranked bands are selected [20], [21]. These methods can be easily implemented, but usually demonstrate limited efficacy in classification tasks [22].

In contrast, subspace-learning-based band selection methods explore the intrinsic structure of HSI to identify optimal band sets, generating band clusters with high intraclass similarity and interclass independence [23]. Assuming that the original high-dimensional data can be reconstructed by several proxies in low-dimensional subspaces [24], more representative features of the original HSI data can be learned [25]. As the quality of the learned coefficient matrix directly impacts the performance of subspace learning, numerous efforts have been made to enhance its accuracy. One effective approach is to embed prior data features or similarity relationships knowledge into the atom matrix. For instance, Sun et al. [26] extended the atom matrix with the unobserved bands to reconstruct original bands in a latent low-rank subspace. Dou et al. [27] introduced an autoencoder to the atom matrix, thereby improving the ability to capture high-order correlations among spectral bands. Besides, in many subspace learning frameworks, the original data matrix itself is typically chosen as the atom matrix used as the subspace basis, known as self-representation

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[28]. Ma and Yang [29] have argued that self-representation is sensitive to the atom matrix, prompting the integration of structural information to refine the subspace basis. In practice, band similarity or spatial similarity [30], [31], [32], has been embedded into the atom matrix to explore the inherent structures of HSI. The improvement of the atom matrix is insufficient in adaptively adjusting the structural properties of the coefficient matrix. Several studies have incorporated additional regularization terms to refine the structural properties of the coefficient matrix [33]. For instance, Shang et al. [34] and Luo et al. [35] introduced graph regularization terms of the coefficient matrix based on the Laplacian matrix derived from original data. By analyzing similar properties of both neighboring and non-neighboring bands, structural regularizations were designed to preserve such relationships explicitly [36]. Li et al. [37] directly imposed the block-diagonal structure on the coefficient matrix to enhance its clustering structure. A spatial–spectral preservation model [38] is developed that leverages the smoothness of the reconstructed data to regularize the coefficient matrix structure. Yan et al. [39] established self-representation modules in both spatial and spectral domains, constructing a unified similarity graph through multiview consistency constraints. These methods provide more sophisticated structural guidance for coefficient matrix learning, thereby enabling more discriminative affinity exploration. Besides, several studies have attempted to characterize band relationships through latent space representations. For example, CapUBS [40] leverages a capsule network to encode multiband features into vector-based capsules and ranks bands according to their latent importance. Similarly, HUTDNet [41] introduces a joint unmixing–detection framework that estimates latent endmembers and abundance maps to characterize material compositions, from which discriminative spectral features can be identified for downstream tasks.

Although the above works demonstrate promising performance by revealing data representability in a low-dimensional space, they still exhibit several limitations. First, after generating the coefficient matrix (denoted as Z), most of the subspace learning methods require an additional manual normalization step to establish the affinity graph $(|Z| + |Z^T|)/2$ between all bands [42], [43]. Such post hoc graph normalization introduces an optimization inconsistency between the representation learning and affinity construction, since the coefficient matrix Z is learned independently of the subsequent normalization operator $(|Z| + |Z^T|)/2$ [44], [45]. The final affinity graph is not jointly optimized with the representation objective, making the representation learning not graph structure-oriented. Thus, this decoupled two-stage procedure implies that the learned optimal Z does not necessarily induce an optimal graph structure [46]. This manual graph normalization may fail to ensure symmetric affinity estimation during the learning process, potentially resulting in fluctuations in affinities and a suboptimal graph structure [47], [48]. Second, these methods usually overlook the mining and optimal reconstruction of affinity structures between pairs of samples, which are crucial for accurate sample grouping.

A doubly stochastic graph is defined as a non-negative and symmetric affinity matrix whose row sums and column sums are all equal to 1. This type of graph inherently unifies graph symmetrization and affinity learning, while preserving the original data connectivity and suppressing spurious or noisy edges [49], [50], [51]. Owing to these favorable properties, doubly stochastic constraints have been increasingly incorporated into subspace learning and clustering models to enhance graph quality recently [52], [53], [54]. For instance, Wang et al. [55] learned a structured doubly stochastic matrix under a low-rank Laplacian constraint. Chen et al. [56] designed a self-representation loss term and combined it with the doubly stochastic coefficient matrix, which could directly generate the cluster indicator. Based on the doubly stochastic property, [57], [58] further imposed idempotent constraints on the similarity graph, making it more consistent with the cluster assignment matrix, for clearer cluster structures exploration. Regarding HSI data processing, the doubly stochastic graph has been used as an initialization graph to improve clustering results in [59]. HPCDL [47] dynamically learned an optimal doubly stochastic graph, rather than simply using it for initialization, which further enhanced the representation of similarity relations. Moreover, Cai et al. [60] extended doubly stochastic graph learning to multimodal clustering by establishing an anchor-based multiview kernel subspace clustering framework, where a unified doubly stochastic graph supported spectral embedding across HSI and LiDAR data.

In general, existing doubly stochastic graph learning methods impose non-negativity, symmetry, and row-sum constraints to normalize the learned graph and maintain meaningful connections while alleviating noise effects, but it poses challenges for optimization [61]. A widely used optimization strategy for doubly stochastic constraints is to separate the constraints and formulate two subproblems via von Neumann successive projection, but this strategy is cumbersome, time-consuming, and challenging to scale [62]. However, directly enforcing multiple constraints, especially when combined with additional idempotent or self-representation constraints, often leads to complex optimization procedures and may limit the scalability of subspace learning models [56], [58].

To overcome the aforementioned limitations, we propose a novel band selection method named implicit doubly stochastic graph-based subspace clustering (IDSGC), which facilitates end-to-end band affinity graph construction without the need for extra graph normalization. The main contributions of IDSGC are summarized as follows.

1) To analyze the similarity relationships within the original HSI, a latent bipartite graph modeling (LBGM) module is designed to decompose the initial graph into the normalized graph. This simple yet effective module reconstructs band affinities in a low-dimensional space, preserving essential pairwise connections while mitigating the impact of noisy links.
2) A novel implicit learning mechanism (ILM) of a doubly stochastic graph is proposed, which establishes the transformation between the coefficient matrix and the band affinity graph. This mechanism maintains the doubly stochastic property on the graph, while simplifying

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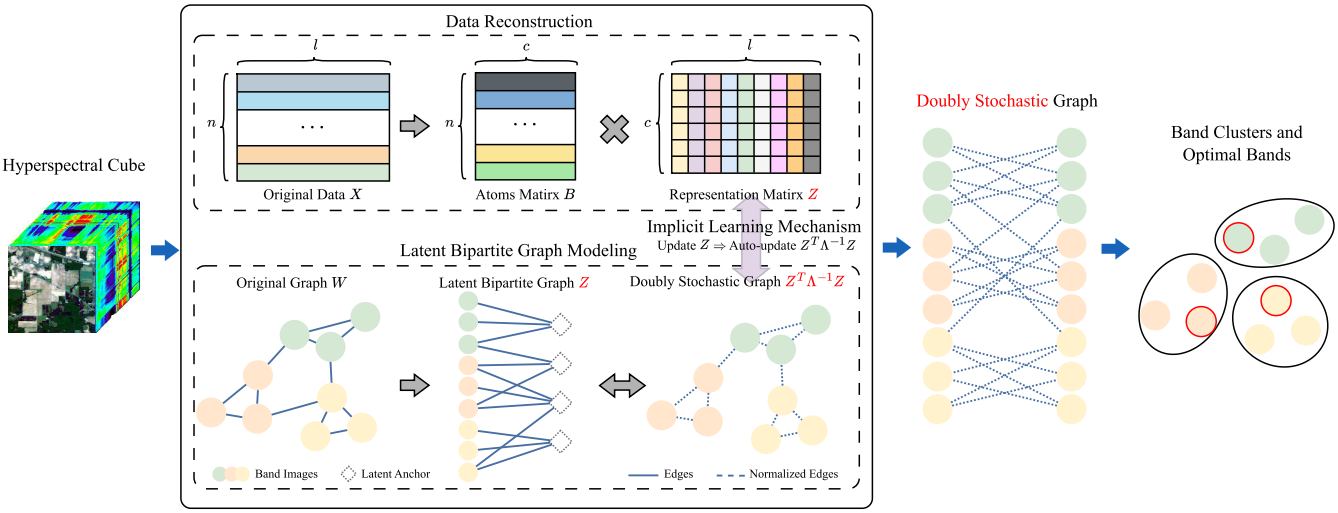


Fig. 1. Flowchart of our work. IDSGC contains two main modules, the LBGM module for affinity preservation and the subspace learning module for data reconstruction. A novel ILM of a doubly stochastic graph naturally unifies these two modules. Thus, our method outputs a doubly stochastic affinity graph (both rows and columns satisfy probability constraint) directly without extra graph normalization.

both the complex constraints imposed on the coefficient matrix and the optimization strategy in doubly stochastic graph learning.

3) By seamlessly integrating the LBGM with subspace learning through ILM, we accomplish collaborative structure regularization of the affinity graph and coefficient matrix under a simple probability constraint. Consequently, in-process normalization is achieved, which directly generates a doubly stochastic graph and eliminates the requirement of manual graph normalization. This method demonstrates enhanced generalization capability across different initial graphs and HSI datasets.

The sections of this article are listed as follows. The detailed description of the proposed model and the optimization strategy is elaborated in Section II. In Section III, we compare the performance of IDSGC with other band selection methods, and provide a detailed analysis on IDSGC model. Finally, Section IV summarizes this article.

II. Methodology

This section presents the details of the IDSGC method, which simultaneously performs data reconstruction and affinity preservation through an ILM, as illustrated in Fig. 1.

A. IDSGC Model Construction

In general, the original HSI cube data is denoted as $H \in \mathbb{R}^{a\times b\times l}$, where a, b, l represent the image length, image width, and the number of spectral bands, respectively. For convenience, the 3-D data is reshaped into a 2-D matrix $X \in \mathbb{R}^{n\times l}$, where each column vector $x_i \in \mathbb{R}^{n\times 1}$ corresponds to a band image, with $n = a \times b$.

Traditional subspace learning aims to reconstruct the original data in a low-dimensional space and obtain a coefficient matrix under certain constraints. The formulation of such a subspace learning framework is as follows:

$$\min_{Z} \mathcal{L}(X - BZ) + \mathcal{C}(Z) \tag{1}$$

where $B \in \mathbb{R}^{n\times c}$ is the dictionary atoms matrix, and $Z \in \mathbb{R}^{c\times l}$ denotes the latent representation in the subspace. The first term, $\mathcal{L}(\cdot)$, measures the reconstruction error between the original HSI data and its embedded representation. The second term, $\mathcal{C}(\cdot)$, imposes constraints on the coefficient matrix Z, typically enforcing a specific structure or sparsity pattern. Based on the learned Z, an affinity graph $(|Z|+|Z^T|)/Z$ is constructed to reflect the pairwise relationships within the original data. However, there are two limitations that need to be addressed: 1) classical subspace learning overlooks preserving the original affinity relations between pairs of samples; and 2) the separate graph symmetrization may lead to suboptimal graph structure.

To preserve the affinities within the original HSI, we enforce reconstruction consistency between the similarity relationships in the original feature space and those in the latent space. Initially, a graph factorization module is considered to exploit affinity structures, formulated as follows:

$$\begin{aligned} \min_{V} \quad & \left\| W - V^T V \right\|_F^2 \\ \text{s.t.} \quad & V \geq 0 \end{aligned} \tag{2}$$

where $W \in \mathbb{R}^{l\times l}$ is the initial graph between original bands. However, this module is sensitive to the initial graph W and is not inherently designed to integrate with the subspace learning module. To overcome these challenges, we leverage the dual interpretability of the coefficient matrix Z. Specifically, the coefficient matrix Z serves both as the underlying representation of the HSI data and as a means of encoding the affinity relationships between the original data and the dictionary atoms. By capitalizing on this dual role, we propose a doubly stochastic transformation of the coefficient matrix Z, leading to the following LBGM module:

$$\begin{aligned} \min_{Z} \quad & \left\| W - Z^T \Lambda^{-1} Z \right\|_F^2 \\ \text{s.t.} \quad & Z^T \mathbf{1} = \mathbf{1}, Z \geq 0 \end{aligned} \tag{3}$$

where $Z \in \mathbb{R}^{c\times l}$ and Λ is a diagonal matrix with $\Lambda_{ii} = \sum_j Z_{ij}$. In the LBGM module, instead of explicitly constructing anchor

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points followed by similarity computation in the original feature space, we model the affinity structure in a latent space spanned by c latent anchors. Under this formulation, the matrix Z is interpreted as the weighted adjacency matrix of a latent bipartite graph, which encodes the relationships between the original spectral bands and the latent anchors. Based on this bipartite structure, the induced band-to-band affinity graph can be naturally reconstructed as $Z^T\Lambda^{-1}Z$. This operation corresponds to projecting the original band relationships onto a lower dimensional latent anchor space, thereby producing a compact and structured graph representation. This module preserves topological affinity properties by decomposing the original graph and clarifying the integration manner with subspace learning. Notably, the LBGM module improves the structural properties of the learned graph $Z^T\Lambda Z$

$$\begin{aligned}\sum_j \left(Z^T\Lambda^{-1}Z\right)_{ij} &= \sum_j\sum_k Z_{ki}\Lambda_{kk}^{-1}Z_{kj} \quad \text{(row sum)}\\ &= \sum_k Z_{ki}\Lambda_{kk}^{-1}\sum_j Z_{kj} = \sum_k Z_{ki} = 1\\ \sum_i \left(Z^T\Lambda^{-1}Z\right)_{ij} &= \sum_i\sum_k Z_{ki}\Lambda_{kk}^{-1}Z_{kj} \quad \text{(column sum)}\\ &= \sum_k Z_{kj}\Lambda_{kk}^{-1}\sum_i Z_{ki} = \sum_k Z_{kj} = 1. \end{aligned} \tag{4}$$

As demonstrated in (4), the learned graph $Z^T\Lambda^{-1}Z$ is a doubly stochastic matrix. This doubly stochastic property provides probabilistic similarity constraints and appropriate normalization of the affinity graph, making it more suitable for measuring the interband affinity.

Combining the LBGM module in (3), we propose a unified model termed IDSGC, formulated as follows:

$$\begin{aligned}&\min_{Z,B} \left\|W - Z^T\Lambda^{-1}Z\right\|_F^2 + \gamma\|X - BZ\|_F^2\\ &\text{s.t. } Z^T\mathbf{1} = \mathbf{1}, Z \geq 0.\end{aligned} \tag{5}$$

It is evident that the two modules in IDSGC are concatenated through the latent representation Z, which reflects its dual role. Specifically, the ILM transforms Z to $Z^T\Lambda^{-1}Z$, and embeds the graph normalization into the learning process itself. The ILM enables the joint optimization of representation learning and affinity construction, thereby eliminating the decoupled two-stage procedure commonly adopted in traditional subspace-based band selection methods. The joint exploration of both the implicit doubly stochastic graph and latent representation enhances the robustness of the learned graph $Z^T\Lambda^{-1}Z$ against the initial graph structures and the presence of noisy edges. As a result, the mutual reinforcement between the two modules leads to a more discriminative characterization of the similarity relationships among band images, thereby improving the band selection performance. To solve IDSGC in (5), an iterative optimization strategy is designed in the following subsection.

B. Optimization Strategy

The proposed IDSGC [see (5)] involves two variables to be optimized, that is, the coefficient matrix Z and dictionary matrix B. We employ the alternating direction method of multipliers (ADMMs) to solve this model, introducing an auxiliary variable J. Consequently, the original problem in (5) and the Lagrangian function are reformulated as follows:

$$\begin{aligned}&\min_{Z,B,J} \left\|W - J^T\Lambda^{-1}Z\right\|_F^2 + \gamma\|X - BZ\|_F^2\\ &\text{s.t. } Z^T\mathbf{1} = \mathbf{1}, Z \geq 0, J = Z\end{aligned} \tag{6}$$

$$\begin{aligned}L(Z,J,B,\Delta,\beta) = &\left\|W - J^T\Lambda^{-1}Z\right\|_F^2 + \gamma\|X - BZ\|_F^2\\ &+ Tr\left(\Delta^T(Z-J)\right) + \beta\|Z-J\|_F^2\end{aligned} \tag{7}$$

where Δ is the Lagrangian multiplier and β is the penalty parameter. The transformed problem is solved by iteratively updating B, J, and Z.

1) *Update B:*

The subproblem for updating B is unconstrained, which is derived from (6)

$$\min_B \|X - BZ\|_F^2. \tag{8}$$

This problem is a standard least squares problem. The optimal solution of B is given by

$$B = XZ^T\left(Z^TZ\right)^{-1}. \tag{9}$$

2) *Update J:*

With B and Z fixed, the Lagrangian function with respect to J can be written as

$$\min_J \left\|W - J^T\hat{Z}\right\|_F^2 + Tr\left(\Delta^T(Z-J)\right) + \beta\|Z-J\|_F^2 \tag{10}$$

where $\hat{Z} = \Lambda^{-1}Z$. This subproblem can be solved by

$$\begin{aligned}&\frac{\partial L_J}{\partial J} = 2\hat{Z}\hat{Z}^TJ + 2\beta I - 2\hat{Z}W^T - \Delta - 2\beta Z = 0\\ &\Rightarrow J = \left(\hat{Z}\hat{Z}^T + \beta I\right)^{-1}\left(\hat{Z}W^T + \beta Z + \frac{\Delta}{2}\right).\end{aligned} \tag{11}$$

3) *Update Z:*

For the optimization of Z, the Lagrangian minimization objective is obtained as follows:

$$\begin{aligned}&\min_Z \left\|W - \hat{J}Z\right\|_F^2 + \gamma\|X - BZ\|_F^2\\ &\qquad + Tr\left(\Delta^T(Z-J)\right) + \beta\|Z-J\|_F^2\\ &\text{s.t. } Z^T\mathbf{1} = \mathbf{1}, Z \geq 0\end{aligned} \tag{12}$$

where $\hat{J} = J^T\Lambda^{-1}$. This subproblem can be reformulated as a matrix trace expression

$$\begin{aligned}&\left\|W - \hat{J}Z\right\|_F^2 + \gamma\|X - BZ\|_F^2\\ &\quad + Tr\left(\Delta^T(Z-J)\right) + \beta\|Z-J\|_F^2\\ &\Leftrightarrow Tr\left(Z^T\hat{J}^T\hat{J}Z + \gamma Z^TB^TBZ + \beta Z^TZ\right.\\ &\qquad \left. + Z^T\Delta - 2Z^T\hat{J}^TW - 2\gamma Z^TB^TX - 2\beta Z^TJ\right)\\ &= Tr\left(Z^TNZ + Z^TM\right)\\ &= \sum_{i=1}^{l}\left(z_i^TNz_i + m_iz_i\right)\end{aligned} \tag{13}$$

where $N = (\hat{J}^T\hat{J} + B^TB + \beta I)$, $M = (\Delta - 2\hat{J}^TW - 2\gamma B^TX - 2\beta J)$. z_i and m_i denote the ith column vectors of matrix Z and M, respectively. Obviously, both the objective and the

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constraints are column-wise separable. Therefore, each column z_i can be optimized independently via

$$\min_{z_i} \ z_i^T N z_i + m_i z_i \quad \text{s.t. } z_i^T \mathbf{1} = \mathbf{1}, z_i \geq 0. \tag{14}$$

Subproblem of z_i: To solve (14), we introduce an auxiliary column vector y

$$\min_{z_i, y} \ z_i^T N y + m_i z_i \quad \text{s.t. } z_i^T \mathbf{1} = \mathbf{1}, z_i \geq 0, z_i = y \tag{15}$$

where the augmented Lagrange method is used to solve the above problem. The corresponding augmented Lagrange function is

$$\min_{z_i, y} \ z_i^T N y + m_i z_i + \frac{\mu}{2} \left\| z_i - y + \frac{\lambda}{\mu} \right\|_2^2 \quad \text{s.t. } z_i^T \mathbf{1} = \mathbf{1}, z_i \geq 0 \tag{16}$$

where λ and μ are the Lagrange multiplier and penalty parameter, respectively.

a) *Update y:*

Fixing z_i, (16) becomes

$$\min_{z_i, y} \ z_i^T N y + \frac{\mu}{2} \left\| z_i - y + \frac{\lambda}{\mu} \right\|_2^2. \tag{17}$$

We can obtain the solution of y by setting the derivative with respect to y to zero

$$y = \frac{1}{\mu} \left(\mu z_i + \lambda - N^T z_i \right). \tag{18}$$

b) *Update z_i:*

Fixing y, (16) becomes

$$\min_{z_i} \ \| z_i - v \|_2^2 \quad \text{s.t. } z_i^T \mathbf{1} = \mathbf{1}, z_i \geq 0 \tag{19}$$

where $v = 1/\mu(\mu y - N y - m - \lambda)$. This problem is a simplex problem under probability constraint, which can be easily solved [63].

c) *Update Lagrange Parameters:*

After optimizing y and z_i, the Lagrange multiplier λ and the penalty parameter μ are updated

$$\lambda = \lambda + \mu (z_i - y) \quad \mu = \min(\rho \mu, \mu_{max}) \tag{20}$$

where ρ is s scaling factor and μ_{max} is an upper bound. Thus, through the augmented Lagrange method with (18)–(20), we solve the column-wise optimization problem (14). We show the pseudo code for solving (14) in Algorithm 1.

Through the update strategies designed for the matrices B, J, and Z, we finally obtain both the optimal representation Z in the subspace and the affinity graph $Z^T \Lambda^{-1} Z$ across all bands. Unlike traditional doubly stochastic graph learning methods, our strategy avoids the reliance on the von Neumann successive projection lemma, which requires repeated projections across multiple subproblems and has quadratic computation and memory costs about the number of samples. In contrast, our strategy decomposes the original matrix-variable problem into a sequence of vector-level subproblems, which simplifies the update rules and facilitates the straightforward incorporation of additional structural constraints on Z. Therefore, the ILM and optimization strategy significantly reduces the computational and storage complexity of the doubly stochastic graph learning, which is linear with the number of samples. With the learned affinity graph, the cluster structure of bands can be explored through spectral analysis as follows:

$$\min_{F} \ Tr\left(F^T Z^T \Lambda^{-1} Z F\right) \quad \text{s.t. } F^T F = I. \tag{21}$$

For the optimal solution of F, eigenvector decomposition (EVD) is performed on $Z^T \Lambda^{-1} Z$. Then, k-means is applied to each row of F to generate cluster labels for each band. Consequently, the bands closest to the cluster centers are selected as the optimal band subset. Algorithm 2 summarizes the whole procedure for solving IDSGC.

Algorithm 1 Algorithm for Solving (14)

Input: $N, m_i, z_i = \mathbf{0}$, $y = z_i$, set $\lambda = 1e^{-7}$, $\mu = 1e^{-2}$, $\mu_{max} = 1e^{7}$, $\rho = 1.5$, $t = 1$.
Output: Optimal vector z_i.

```
while not converged do
    Update y^t via Eq.(18);
    Update z_i^t through solving Eq.(19);
    Update λ^{t+1}, μ^{t+1} through Eq.(20);
    t = t + 1.
end
```

Algorithm 2 Algorithm for Solving IDSGC (5)

Input: Hyperspectral image X, optimal band number c, initial graph W, and balanced parameter γ.
Output: Optimal band subset.

```
Randomly initialize coefficient matrix Z satisfying the row sum constraint;
while not converged do
    Update B via Eq.(9);
    Update J via Eq.(11);
    for i = 1 : l do
        Update each column of z_i by Algorithm 1;
    end
    Δ = Δ + β(Z − J);
    β = min(ρβ, β_max);
end
Perform EVD on graph Z^T Λ^{-1} Z to obtain cluster indicator F;
Perform k-means on each row of F to obtain cluster label of each band;
Select the band nearest to cluster center in each class into the optimal band subset.
```

C. Model Analyses

1) Compared With Traditional Subspace Clustering: As shown in (1), traditional subspace clustering methods generally

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consist of a reconstruction term for representation learning and a regularization term imposed on the coefficient matrix Z. In these methods, the affinity graph is typically constructed through a posthoc symmetrization operation, i.e., $(|Z|+|Z^T|)/2$. Such a construction is decoupled from the optimization objective and relies on a nonsmooth mapping with respect to Z, meaning that the graph structure is not directly constrained during the learning process.

To enable a principled comparison, we rewrite both the traditional methods and the proposed IDSGC into a unified optimization form

$$\min_{Z} \mathcal{L}(X - BZ) + \mathcal{C}_Z(Z) \tag{22}$$

where the essential difference lies in the definition of the regularization term $\mathcal{C}_Z(Z)$.

In the proposed model, the regularization term is defined as $\mathcal{C}_Z(Z) = \|W - Z^T\Lambda^{-1}Z\|_F^2$, which enforces direct alignment between the learned graph and the initial affinity graph W. This term preserves the original affinity structure effectively that traditional subspace clustering from the following perspectives.

1) *Implicit Symmetrization via Structure-Constrained Optimization:*
   Unlike the explicit symmetrization $(|Z| + |Z|^T)/2$ used in traditional methods, the affinity graph in IDSGC is parameterized as $G(Z) = Z^T\Lambda^{-1}Z$. For any feasible Z, $G(Z)$ satisfies

$$G(Z) = G(Z)^T, \quad G(Z) \geq 0$$

   thereby guaranteeing symmetry and non-negativity intrinsically during optimization. Therefore, the structural properties of the affinity matrix are embedded into the parameterization itself, rather than imposed through an external postprocessing operation.
   Moreover, since $G(Z)$ is quadratic in Z, the mapping from Z to the affinity graph is continuously differentiable within the feasible domain, which enables stable and joint optimization of representation learning and graph construction. By contrast, the absolute-value-based symmetrization adopted in traditional subspace clustering methods is nondifferentiable at zero, which may yield ambiguous gradients and prevent fully coupled end-to-end optimization of the affinity structure.
2) *Latent Bipartite Graph Decomposition of the Affinity Structure:* The factorization $W \approx Z^T\Lambda^{-1}Z$ decomposes the original affinity graph into a latent bipartite structure between data samples and latent anchors encoded by Z. Under the constraint $Z^T\mathbf{1} = \mathbf{1}$, the reconstructed graph $Z^T\Lambda^{-1}Z$ lies in the space of doubly stochastic matrices, which forms a structured subset of general affinity matrices.
   Consequently, minimizing $\|W - Z^T\Lambda^{-1}Z\|_F^2$ corresponds to projecting the original graph onto a structure-constrained matrix manifold, thus preserving the dominant affinity patterns of W while suppressing inconsistent or noisy connections. Therefore, the proposed objective function ensures preservation of the original affinity structure not only by minimizing the discrepancy to W, but also by embedding structural constraints directly into the optimization process, leading to a more tightly coupled graph learning formulation.

2) Graph Initialization Manner: An initial graph is the input of our IDSGC method in Algorithm 2, and four different graph initialization manners are shown as follows.

1) *Effective Neighbor Assignment [64] (Denote as Graph1):* This manner constructs graph by solving

$$\min_{w_i^T\mathbf{1}=1, w_i\geq 0} \sum h(x_i, x_j) W_{ij} + \lambda \sum W_{ij}^2 \tag{23}$$

   where $h = \|x_i - x_j\|_2^2$. The closed-from solution is

$$W_{ij} = \frac{h(x_i, x_{k+1}) - h(x_i, x_j)}{\sum_{j'=1}^{k}\left[h(x_i, x_{k+1}) - h(x_i, x_{j'})\right]}. \tag{24}$$

   In the above solution, k is a manual parameter. As the constructed is the initial graph and our method is robust to it, we just set $k = 3$ according to [64].
2) *ε-Neighborhood Graph (Denote as Graph2):* If the distance between two points x_i, x_j is smaller than ε, we connect them with the edge weight $W_{ij} = W_{ji} = exp(-h/2\sigma^2)$, σ is a parameter.
3) *k-Neighborhood Graph (Denote as Graph3):* If points x_j belongs to the k-nearest neighbors of x_i, we connect two points with $W_{ij} = exp(-h/2\sigma^2)$.
4) *Fully Connected Graph (Denote as Graph4):* For any two pairs of points, let $W_{ij} = exp(-h/2\sigma^2)$.

In these four manners, only the integer parameter k needs to be tuned for Graph1, while the other involve setting the value of σ. Therefore, we use Graph1 as the initial graph of our method, with k set to 5, as suggested in their work [64]. The other three approaches are used as the comparison graph initializations in the ablation studies to validate the robustness of the IDSGC against different initial graph configurations.

3) Convergence Analysis: In the proposed method, the objective function (5) is non-negative, and three variables B, J and Z are needed to be solved. An alternating optimization strategy is adopted to iteratively update each variable while fixing the others, thereby ensuring that the objective function value decreases monotonically at each step. As a result, the iterative process guarantees that the objective value converges to a local minimum. Therefore, it can be concluded that Algorithm 2 converges to a locally optimal solution of the IDSGC model in (5).

4) Complexity Analysis: According to Algorithm 2, the computational efficiency of our algorithm is influenced by three main factors. Updating B requires $\mathcal{O}(l^3)$ to compute the matrix inverse Z^TZ and $\mathcal{O}(nlc)$ to compute B. The update of J takes $\mathcal{O}(2lc^2 + l^2c + c^3)$. The optimization of Z is performed in a column-wise manner, with a complexity $\mathcal{O}(lt)$, where t is the iteration number in Algorithm 1. Therefore, the overall complexity of our IDSGC model in Algorithm 2 is $\mathcal{O}(T(l^3 + nlc + 2lc^2 + l^2c + c^3 + lt))$, where T represents the total number of iterations. In practice, since $c \ll l \ll n$, the dominant term of algorithm complexity is $\mathcal{O}(T(l^3 + nlc))$.

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III. Experiments

In this section, we begin with the experimental setup, which covers the HSI datasets, the compared methods, the classifiers, and the evaluation metrics. Next, we perform comparative experiments to assess the effectiveness of IDSGC against other band selection methods. We then analyze the convergence behavior and parameter sensitivity of the IDSGC method. Finally, we conduct ablation studies to demonstrate the advantages of IDSGC.

A. Experimental Setup

The experiments are conducted with MATLAB 2021b on a computer with a 3.2-GHz CPU and 64 GB of RAM. To evaluate the performance of our band selection method, we conduct classification experiments on four HSI datasets, comparing it with eight other band selection methods. Detailed descriptions of the HSI datasets, comparison methods, and classification configurations are provided below.

1) HSI Datasets:

1) *Houston:* This dataset is acquired using the CASI-1500 sensor, with a spatial resolution of 2.5 m and 145 spectral bands covering wavelengths from 380 to 1050 nm. It contains 15 land-cover classes, and each image has dimensions of 349×1905 pixels.
2) *IndianPines:* This dataset is acquired using the AVIRIS sensor, with a spatial resolution of 20 m and 220 spectral bands covering wavelengths from 400 to 2500 nm. Due to the water absorption, 20 bands are discarded, and the remaining 200 bands are used in our experiments. It contains 16 land-cover classes (we use nine classes with the largest sample size here), and each image has dimensions of 145×145 pixels.
3) *PaviaUniversity (PaviaU):* This dataset is acquired using the ROSIS sensor, with a spatial resolution of 1.3 m and 103 spectral bands covering wavelengths from 430 to 860 nm. It contains nine land-cover classes, and each image has dimensions of 610×340 pixels.
4) *Botswana:* This dataset is acquired using the NASA EO-1 sensor, with a spatial resolution of 30 m and 145 spectral bands covering wavelengths from 400 to 2500 nm. It contains 14 land-cover classes, and each image has dimensions of 1476×256 pixels.
5) *WHU-Hi-LongKou:* This dataset is acquired using an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro UAV platform, with a spatial resolution of 0.463 m and 270 spectral bands covering wavelengths 400 to 1000 nm. It contains nine land-cover classes, and each image has dimensions of 550×400 pixels.
6) *QUH-Tangdaowan:* This dataset is acquired using a Gaiasky mini2-VN imaging spectrometer equipped on a DJI Matrice 600 Pro UAV platform, with a spatial resolution of 0.15 m and 176 spectral bands covering wavelengths 400–1000 nm. It contains 18 land-cover classes, and each image has a dimension of 1740×860 pixels.

The detailed land-cover categorizations of the used datasets are listed in Table I.

2) Compared Methods:

1) *MGSR [30]:* A graph self-representation-based method. This method utilizes the structural information within spatial regions to refine self-representation atoms and learns a weighted matrix for band ranking.
2) *RDGSR [31]:* A graph self-representation-based method. This method utilizes both spatial and spectral structural information to enhance the self-representation of atoms and generates a weighted matrix for band ranking.
3) *HLFC [65]:* A clustering-based method. This method first extracts latent feature graphs from different superpixel regions and then performs spectral clustering with a graph fusion strategy.
4) *RFGEC [23]:* A subspace learning-based method. This method integrates a graph filter with subspace clustering to learn the data representation, from which a band affinity graph is constructed.
5) *TGSR [32]:* A graph self-representation-based method. This method constructs global and local graphs across different superpixel regions, and learns a weighted matrix of each region under an overall Schatten p-norm constraint.
6) *SLFLRSC [33]:* A graph self-representation-based method. This method explores the latent features of each superpixel region and establishes a low-rank subspace learning model to maintain the structural smoothness and index adjacency.
7) *GPCBS [66]:* A global partition clustering-based method. This method first partitions the bands into several ordered subgroups and then selects a representative band from each group.
8) *DVS3C [39]:* A subspace learning-based method. This method treats spatial and spectral information as distinct views and learns a unified coefficient matrix across all views based on self-representation.

Among the aforementioned methods, HLFC and GPCBS can directly learn the cluster indicators for all bands. In contrast, other band selection methods first learn a coefficient matrix with self-representation and then apply different strategies to identify band clusters or an optimal band subset. The MGSR and RDGSR methods treat the coefficient matrix as a weighted matrix for each band and rank the bands by calculating the ℓ_2-norm of each row. The RFGEC method performs spectral clustering directly on the learned coefficient matrix. Meanwhile, the TGSR, SLFLRSC, and DVS3C methods construct a symmetric graph via $(|Z|+|Z^T|)/2$, followed by spectral clustering on the symmetric graph. We summarize the postprocessing types and manners used by these comparative methods in Table II.

3) Classification Settings: To evaluate the effectiveness of the optimal band subsets, classification experiments are conducted using k-nearest neighborhood (KNN), support vector machine (SVM) classifiers, and random forest (RF). For both supervised classifiers, 10% of the pixels are randomly selected for training, while the remaining 90% pixels are reserved for testing. The parameter k in KNN is set to 5. For SVM

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TABLE I

DETAILS OF EACH LAND-COVER CATEGORIZATION OF THE USED DATASETS IN EXPERIMENTS

No.	Houston		IndianPines		PaviaU		Botawana		WHU-Hi-LongKou		QUH-Tangdaowan	
	Class Name	Pixels	Class Name	Pixels	Class Name	Pixels	Class Name	Pixels	Class Name	Pixels	Class Name	Pixels
1	Healthy Grass	1251	Corn-notill	1428	Asphalt	6631	Water	270	Corn	34511	Rubber Track	25849
2	Stressed Grass	1254	Corn-mintill	830	Meadows	18649	Hippo grass	101	Cotton	8374	Flagging	55553
3	Synthetic Grass	697	Grass-pasture	483	Gravel	2099	Floodplain grasses1	251	Sesame	3031	Sandy	34037
4	Tree	1244	Grass-trees	730	Trees	3064	Floodplain grasses2	215	Broad-leaf soybean	63212	Asphalt	60690
5	Soil	1242	Hay-windrowed	478	Painted metal sheets	1345	Reeds	269	Narrow-leaf soybean	4151	Boardwalk	1862
6	Water	325	Soybean-notill	972	Bare soil	5029	Riparian	269	Rice	11854	Rocky shallows	37125
7	Residential	1268	Soybean-mintill	2455	Bitumen	1330	Firescar2	259	Water	67056	Grassland	14127
8	Commercial	1244	Soybean-clean	593	Self-blocking bricks	3682	Island inlerior	203	Roads and houses	7124	Bulrush	64087
9	Road	1252	Woods	1265	Shadows	947	Acacia woodlands	314	Mixed weed	5229	gravel road	30695
10	Highway	1227					Acacia shrub lands	248			Ligustrum vicaryi	1783
11	Railway	1235					Acacia grasslands	305			Coniferous pine	21236
12	Parking Lot1	1233					Short mopani	181			Spiraea	749
13	Parking Lot2	469					Mixcd mopani	268			Bare soil	1686
14	Tennis Court	428					Expose soils	95			Buxus sinica	886
15	Running Track	660									Photinia serrulata	14020
16											Populus	140904
17											Ulmus pumila L	9802
18											Seawater	42275
Total		15029		9234		42776		3248		204542		557366

TABLE II

POSTPROCESSING TYPES AND MANNERS OF THE COMPETITIVE METHODS ON THE LEARNED COEFFICIENT MATRIX

Method	Post-Processing Types	Post-Processing Manners
MGSR	Ranking	ℓ_2-norm
RDGSR	Ranking	ℓ_2-norm
RFGEC[1]	Clustering	\
TGSR	Clustering	Symmetric Graph
SLFLRSC	Clustering	Symmetric Graph
DVS3C	Clustering	Symmetric Graph

[1] RFGEC needs none extra processing.
[2] HLFC learns cluster indicator directly.
[3] GPCBS is an ordered partition method without coefficient matrix.

classification, we employ the LIBSVM tools with a linear kernel function, and the optimal cost and gamma parameters are determined during the training process. The number of trees in RF is set to 20.

To ensure the statistical reliability of our experiments, each classification trial is repeated ten times. Classification performance is evaluated using three metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (Kappa). OA measures the percentage of pixels that have been assigned to the correct category, AA reflects the mean classification accuracy per land-cover class, and Kappa quantifies the consistency between the predicted labels and ground truth.

B. Comparative Results and Analysis

In the comparative experiments, we test the classification results using the optimal band subsets selected by all methods. We set the selected bands number varying from 3 to 30, with an interval of three bands. The classification results are presented in Tables III–V and Fig. 2.

To evaluate the optimal performance of each method within our experimental setup, we record the highest performance metrics attained by each method in Tables III–V. These tables display the highest OA, AA, and Kappa values, and the per-class accuracy by KNN. For the IDSGC method, we record the corresponding number of bands and value of γ in these tables. From the results, several observations can be obtained:

1) In Table II, the competitors are categorized into two main groups based on the types of postprocessing applied to the learned coefficient matrix (if applicable). From the experimental results, it can be considered that the clustering-based postprocessing methods tend to yield band subsets of superior quality compared to those obtained through ranking-based approaches. This performance difference arises because the clustering strategy considers both the similarity and independence among bands, thereby promoting a more representative and discriminative subset. Moreover, our IDSGC method outperforms all clustering-based postprocessing methods. This can be attributed to our ILM of a doubly stochastic graph, which performs in-process graph normalization, thus alleviating the issue of suboptimal affinity estimation that results from separate normalization steps.
2) In most cases, the number of bands corresponding to the highest value of all three metrics is consistent for a given method. However, certain exceptions are observed. For example, inconsistencies occur in the following cases: MGSR on IndianPines and PaviaU, RDGSR on Botswana, HLFC on IndianPines and Botswana, RFGEC on Botswana, TGSR on IndianPines and Botswana, SLFLRSC on IndianPines and WHU-Hi-LongKou, GPCBS on Houston and WHU-Hi-LongKou, and DVS3C on IndianPines and PaviaU. Furthermore, the number of bands yielding the highest AA often differs from those that yield the optimal OA and Kappa. As described in the classification settings, each metric emphasizes different aspects of classification performance. The discrepancy in the number of bands required to achieve the highest values for each metric declares that the selected band subset may be suboptimal, as it fails to simultaneously maximize accuracy across all pixels and within individual categories. In contrast, the proposed IDSGC method consistently achieves the highest performance across all three metrics with the same number of bands, regardless of the HSI datasets or classifiers used.

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TABLE III

PER CLASS AA, OA, AA, AND KAPPA ON HOUSTON, INDIANPINES, AND PAVIAU DATASETS (MEAN ± STD). FOR THE COMPARED METHODS, THE RELATED BAND NUMBERS ARE RECORDED IN PARENTHESES. FOR OUR METHOD, THE RELATED BAND NUMBER AND PARAMETER γ VALUE ARE RECORDED IN PARENTHESES. THE MAXIMUM VALUES ARE MARKED IN BOLD

Data	Classifier	Metric	MGSR(2022)	RDGSR(2022)	HLFC(2023)	RFGEC(2024)	TGSR(2024)	SLFLRSC(2024)	GPCBS(2025)	DVS3C(2025)	Our
Houston	KNN	1	97.07 ± 0.86	97.12 ± 1.12	98.28 ± 1.26	96.86 ± 0.78	96.18 ± 1.09	96.34 ± 0.99	$\mathbf{97.76 \pm 0.29}$	97.27 ± 1.18	96.96 ± 1.85
		2	98.46 ± 0.74	97.50 ± 1.63	$\mathbf{99.40 \pm 0.98}$	96.60 ± 1.27	97.06 ± 1.14	98.02 ± 0.89	97.59 ± 0.80	98.38 ± 0.56	96.16 ± 0.84
		3	96.05 ± 0.96	97.72 ± 0.59	$\mathbf{99.12 \pm 0.68}$	96.53 ± 0.99	97.30 ± 0.96	97.55 ± 0.92	97.47 ± 0.93	97.65 ± 0.60	96.66 ± 0.98
		4	97.30 ± 0.80	97.71 ± 0.87	$\mathbf{98.39 \pm 0.63}$	95.57 ± 1.39	97.77 ± 1.22	97.38 ± 0.22	96.18 ± 2.27	97.72 ± 0.96	97.38 ± 0.41
		5	97.84 ± 1.87	98.23 ± 0.67	$\mathbf{98.97 \pm 0.76}$	98.05 ± 0.53	98.62 ± 0.12	98.59 ± 0.25	98.25 ± 0.36	98.81 ± 0.14	98.39 ± 0.68
		6	89.62 ± 3.38	$\mathbf{93.43 \pm 1.26}$	91.11 ± 5.61	85.05 ± 4.61	89.54 ± 5.07	91.06 ± 3.05	89.17 ± 5.14	86.63 ± 4.84	88.40 ± 5.15
		7	77.83 ± 1.96	79.70 ± 1.69	82.31 ± 4.24	79.40 ± 3.62	78.72 ± 1.27	78.88 ± 1.38	79.23 ± 3.10	81.36 ± 3.79	$\mathbf{83.19 \pm 3.44}$
		8	71.05 ± 2.02	74.91 ± 1.90	79.41 ± 4.21	77.87 ± 0.72	79.35 ± 2.66	75.34 ± 5.44	79.02 ± 2.34	77.99 ± 3.42	$\mathbf{79.62 \pm 2.20}$
		9	72.14 ± 1.44	73.93 ± 2.13	77.04 ± 1.07	73.04 ± 2.32	75.03 ± 3.46	77.21 ± 2.18	74.86 ± 2.78	77.44 ± 1.76	$\mathbf{78.15 \pm 1.25}$
		10	78.90 ± 2.46	84.96 ± 1.42	85.39 ± 2.01	84.47 ± 1.76	$\mathbf{85.77 \pm 2.92}$	$\mathbf{85.77 \pm 1.36}$	84.07 ± 1.76	84.81 ± 2.07	84.74 ± 2.51
		11	68.36 ± 4.35	73.58 ± 3.07	76.97 ± 2.50	76.69 ± 1.71	77.52 ± 1.23	74.08 ± 3.66	75.13 ± 2.42	73.25 ± 3.18	$\mathbf{80.38 \pm 2.09}$
		12	56.65 ± 2.30	70.72 ± 1.76	69.79 ± 5.32	70.79 ± 1.50	71.19 ± 3.00	70.76 ± 1.97	70.31 ± 2.98	71.00 ± 1.22	$\mathbf{76.85 \pm 1.59}$
		13	4.59 ± 1.75	9.06 ± 1.55	10.89 ± 2.43	9.60 ± 2.62	10.52 ± 3.30	7.61 ± 2.90	8.11 ± 2.42	7.62 ± 2.04	$\mathbf{11.96 \pm 3.06}$
		14	90.10 ± 1.93	92.88 ± 2.02	96.31 ± 1.12	93.83 ± 2.38	94.18 ± 2.16	94.72 ± 1.00	92.71 ± 3.80	93.69 ± 1.82	$\mathbf{96.42 \pm 1.72}$
		15	97.14 ± 0.82	96.38 ± 1.83	$\mathbf{98.72 \pm 0.25}$	98.01 ± 0.36	97.83 ± 0.79	97.34 ± 0.44	97.70 ± 1.16	97.32 ± 0.97	97.71 ± 0.81
		OA	81.01 ± 0.39 (30)	84.15 ± 0.85 (24)	85.97 ± 0.35 (27)	83.83 ± 0.57 (24)	84.79 ± 0.38 (12)	84.23 ± 0.32 (21)	84.32 ± 0.32 (30)	84.49 ± 0.32 (27)	$\mathbf{86.05 \pm 0.62}$ (30, 0.1)
		AA	79.52 ± 0.54 (30)	82.52 ± 0.93 (24)	84.14 ± 0.54 (27)	82.16 ± 0.57 (24)	83.10 ± 0.43 (12)	82.71 ± 0.18 (21)	82.50 ± 0.29 (24)	82.73 ± 0.32 (27)	$\mathbf{84.20 \pm 0.63}$ (30, 0.1)
		Kappa	79.95 ± 0.41 (30)	83.21 ± 0.88 (24)	85.10 ± 0.37 (27)	82.88 ± 0.61 (24)	83.99 ± 0.39 (12)	83.30 ± 0.33 (21)	83.39 ± 0.33 (30)	83.56 ± 0.33 (27)	$\mathbf{85.19 \pm 0.64}$ (30, 0.1)
	SVM	OA	79.19 ± 1.39 (30)	82.40 ± 0.74 (30)	84.01 ± 0.62 (27)	84.06 ± 1.47 (30)	85.14 ± 1.53 (24)	85.26 ± 1.47 (30)	84.55 ± 2.12 (30)	85.05 ± 1.41 (30)	$\mathbf{85.76 \pm 0.99}$ (30, 1)
		AA	78.83 ± 1.11 (30)	81.71 ± 0.64 (30)	83.58 ± 1.10 (27)	83.98 ± 1.33 (30)	85.12 ± 1.17 (24)	84.95 ± 1.33 (30)	84.47 ± 1.95 (30)	85.01 ± 1.46 (30)	$\mathbf{85.99 \pm 0.91}$ (30, 1)
		Kappa	78.09 ± 1.39 (30)	81.40 ± 0.76 (30)	83.08 ± 1.36 (27)	83.14 ± 1.52 (30)	84.26 ± 1.58 (24)	84.38 ± 1.53 (30)	83.64 ± 2.20 (30)	84.17 ± 1.47 (30)	$\mathbf{84.91 \pm 1.03}$ (30, 1)
	RF	OA	81.52 ± 0.34 (30)	84.81 ± 0.17 (30)	86.57 ± 0.23 (24)	85.92 ± 0.60 (30)	86.42 ± 0.38 (12)	86.63 ± 0.26 (18)	86.57 ± 0.48 (30)	86.87 ± 0.62 (30)	$\mathbf{87.33 \pm 0.47}$ (21, 1000)
		AA	80.32 ± 0.28 (30)	83.64 ± 0.31 (30)	85.24 ± 0.32 (24)	84.44 ± 0.69 (30)	85.23 ± 0.22 (12)	85.19 ± 0.70 (18)	85.30 ± 0.56 (30)	85.40 ± 0.20 (27)	$\mathbf{86.12 \pm 0.34}$ (21, 1000)
		Kappa	80.48 ± 0.35 (30)	83.90 ± 0.17 (30)	85.74 ± 0.24 (24)	85.06 ± 0.62 (30)	85.59 ± 0.40 (12)	85.80 ± 0.27 (18)	85.74 ± 0.50 (30)	86.05 ± 0.65 (30)	$\mathbf{86.53 \pm 0.49}$ (21, 1000)
IndianPines	KNN	1	46.16 ± 3.50	60.57 ± 2.58	54.06 ± 2.52	61.52 ± 2.09	62.15 ± 3.06	59.36 ± 2.79	60.99 ± 2.01	64.20 ± 2.77	$\mathbf{66.75 \pm 1.44}$
		2	42.11 ± 4.17	50.57 ± 1.06	46.96 ± 2.74	47.45 ± 2.41	52.18 ± 2.12	54.46 ± 1.60	45.83 ± 2.25	37.75 ± 3.19	$\mathbf{54.67 \pm 1.55}$
		3	70.15 ± 5.67	$\mathbf{79.21 \pm 3.83}$	70.27 ± 3.34	75.83 ± 3.91	73.43 ± 2.71	73.18 ± 2.75	75.21 ± 1.97	78.34 ± 4.96	68.32 ± 2.14
		4	94.69 ± 0.69	96.49 ± 0.40	95.80 ± 0.92	95.90 ± 1.69	98.14 ± 0.41	96.78 ± 1.55	96.58 ± 0.74	$\mathbf{97.72 \pm 0.37}$	97.60 ± 0.89
		5	95.30 ± 1.07	98.78 ± 1.04	97.05 ± 0.97	98.53 ± 0.35	96.75 ± 1.70	97.94 ± 1.00	99.02 ± 0.86	99.12 ± 0.30	$\mathbf{99.81 \pm 0.10}$
		6	36.90 ± 3.52	49.50 ± 2.62	67.76 ± 5.02	70.68 ± 2.84	73.37 ± 1.01	75.16 ± 3.22	71.67 ± 2.95	65.85 ± 2.26	$\mathbf{75.31 \pm 1.16}$
		7	62.22 ± 2.74	69.79 ± 2.90	67.29 ± 2.50	73.20 ± 1.33	71.38 ± 3.12	73.59 ± 4.28	70.85 ± 1.52	70.78 ± 1.06	$\mathbf{75.44 \pm 1.85}$
		8	14.33 ± 1.43	32.24 ± 1.93	29.78 ± 5.16	31.62 ± 4.31	35.58 ± 5.00	30.95 ± 6.12	30.59 ± 5.24	21.84 ± 6.02	$\mathbf{36.93 \pm 6.07}$
		9	94.28 ± 3.20	97.20 ± 0.32	88.85 ± 1.87	94.99 ± 1.68	93.87 ± 1.94	$\mathbf{98.03 \pm 0.45}$	96.53 ± 0.84	94.82 ± 2.32	96.91 ± 0.79
		OA	61.28 ± 0.61 (27)	70.06 ± 0.45 (30)	68.49 ± 0.62 (9)	72.46 ± 0.85 (27)	73.55 ± 0.74 (24)	73.28 ± 1.26 (30)	71.86 ± 0.63 (30)	70.42 ± 0.36 (18)	$\mathbf{75.24 \pm 0.74}$ (27, 0.1)
		AA	61.79 ± 0.91 (27)	70.48 ± 0.45 (30)	68.65 ± 0.63 (9)	72.19 ± 0.61 (27)	72.98 ± 0.79 (21)	73.27 ± 1.45 (24)	71.92 ± 0.59 (30)	70.05 ± 0.30 (30)	$\mathbf{74.64 \pm 0.63}$ (27, 0.1)
		Kappa	58.33 ± 0.59 (27)	67.37 ± 0.48 (30)	65.77 ± 0.61 (9)	69.85 ± 0.81 (27)	71.01 ± 0.77 (24)	71.29 ± 1.29 (30)	69.28 ± 0.63 (30)	67.74 ± 0.36 (18)	$\mathbf{72.75 \pm 0.77}$ (27, 0.1)
	SVM	OA	66.16 ± 0.79 (21)	68.95 ± 0.73 (21)	72.38 ± 1.14 (27)	77.78 ± 0.73 (30)	77.04 ± 0.44 (24)	76.63 ± 0.48 (18)	78.06 ± 0.17 (30)	72.13 ± 0.38 (27)	$\mathbf{78.49 \pm 0.67}$ (30, 0.1)
		AA	66.07 ± 1.12 (21)	70.46 ± 0.97 (21)	72.87 ± 1.41 (27)	78.66 ± 1.12 (30)	77.96 ± 1.31 (30)	78.24 ± 0.61 (18)	78.73 ± 0.32 (30)	72.45 ± 0.90 (27)	$\mathbf{79.37 \pm 0.67}$ (30, 0.1)
		Kappa	63.13 ± 0.84 (21)	66.14 ± 0.82 (21)	69.83 ± 1.21 (27)	75.49 ± 0.79 (30)	74.66 ± 0.49 (24)	74.19 ± 0.51 (18)	75.76 ± 0.17 (30)	69.47 ± 0.44 (27)	$\mathbf{76.24 \pm 0.70}$ (30, 0.1)
	RF	OA	66.74 ± 0.60 (27)	71.08 ± 0.62 (30)	72.67 ± 0.92 (27)	75.64 ± 0.54 (30)	76.07 ± 0.12 (27)	76.20 ± 0.62 (12)	76.36 ± 0.38 (30)	73.97 ± 0.94 (27)	$\mathbf{77.31 \pm 0.54}$ (30, 10)
		AA	66.48 ± 0.90 (24)	70.42 ± 1.15 (30)	71.70 ± 0.75 (9)	74.67 ± 0.43 (30)	74.90 ± 0.65 (27)	75.83 ± 0.60 (18)	75.77 ± 0.66 (30)	73.64 ± 0.79 (27)	$\mathbf{76.64 \pm 0.48}$ (30, 10)
		Kappa	63.76 ± 0.60 (27)	68.30 ± 0.67 (30)	69.94 ± 0.97 (27)	73.09 ± 0.54 (30)	73.54 ± 0.15 (27)	73.72 ± 0.66 (12)	73.88 ± 0.38 (30)	71.36 ± 0.95 (27)	$\mathbf{74.86 \pm 0.54}$ (30, 10)
PaviaU	KNN	1	86.21 ± 0.73	85.82 ± 0.87	87.07 ± 1.24	86.58 ± 0.61	$\mathbf{88.05 \pm 1.01}$	87.58 ± 0.42	87.40 ± 0.30	86.08 ± 0.59	87.48 ± 1.05
		2	96.70 ± 0.26	96.54 ± 0.61	97.22 ± 0.30	97.40 ± 0.29	97.03 ± 0.45	97.41 ± 0.20	96.70 ± 0.20	97.53 ± 0.11	$\mathbf{97.73 \pm 0.19}$
		3	58.81 ± 1.87	58.47 ± 1.55	68.11 ± 3.29	68.55 ± 2.23	67.68 ± 2.25	64.37 ± 2.19	65.92 ± 1.08	62.96 ± 2.94	$\mathbf{71.44 \pm 1.64}$
		4	73.94 ± 1.78	82.62 ± 0.66	81.74 ± 1.09	74.80 ± 1.24	$\mathbf{83.26 \pm 0.73}$	80.00 ± 0.55	80.79 ± 1.94	79.65 ± 2.18	79.64 ± 1.84
		5	93.54 ± 1.20	$\mathbf{99.16 \pm 0.10}$	99.03 ± 0.21	99.10 ± 0.24	99.05 ± 0.13	99.00 ± 0.19	99.07 ± 0.12	98.69 ± 0.27	99.12 ± 0.14
		6	62.71 ± 1.75	58.17 ± 1.70	58.56 ± 0.62	54.38 ± 1.03	58.41 ± 1.20	56.34 ± 1.02	55.74 ± 1.34	56.81 ± 1.41	$\mathbf{64.54 \pm 1.89}$
		7	83.24 ± 2.23	74.93 ± 1.86	81.42 ± 2.87	$\mathbf{84.02 \pm 0.80}$	81.93 ± 2.44	81.41 ± 3.95	80.68 ± 1.70	81.35 ± 2.30	83.81 ± 1.82
		8	81.83 ± 1.15	81.20 ± 0.60	81.42 ± 1.50	83.23 ± 1.83	83.64 ± 0.68	83.53 ± 1.29	$\mathbf{84.67 \pm 1.03}$	83.25 ± 1.12	84.43 ± 0.81
		9	99.88 ± 0.00	99.74 ± 0.05	99.84 ± 0.06	99.71 ± 0.10	99.85 ± 0.05	99.82 ± 0.10	99.71 ± 0.01	99.85 ± 0.10	$\mathbf{99.98 \pm 0.05}$
		OA	85.88 ± 0.18 (30)	85.73 ± 0.27 (30)	87.02 ± 0.13 (18)	86.07 ± 0.15 (30)	87.06 ± 0.15 (27)	86.59 ± 0.16 (30)	86.38 ± 0.11 (18)	86.35 ± 0.18 (30)	$\mathbf{88.08 \pm 0.23}$ (27, 0.1)
		AA	81.88 ± 0.31 (30)	81.85 ± 0.36 (30)	84.10 ± 0.30 (18)	83.08 ± 0.30 (30)	84.32 ± 0.31 (27)	83.27 ± 0.24 (30)	83.41 ± 0.24 (18)	82.91 ± 0.34 (24)	$\mathbf{85.35 \pm 0.36}$ (27, 0.1)
		Kappa	82.08 ± 0.20 (30)	82.86 ± 0.32 (30)	83.45 ± 0.16 (18)	82.27 ± 0.20 (30)	83.52 ± 0.20 (27)	82.90 ± 0.20 (30)	82.66 ± 0.14 (18)	82.59 ± 0.23 (30)	$\mathbf{84.74 \pm 0.29}$ (27, 0.1)
	SVM	OA	85.22 ± 0.12 (30)	86.12 ± 0.33 (30)	88.91 ± 0.31 (27)	88.87 ± 0.18 (30)	89.16 ± 0.17 (27)	89.00 ± 0.17 (30)	89.12 ± 0.11 (30)	89.11 ± 0.17 (27)	$\mathbf{89.96 \pm 0.09}$ (30, 0.1)
		AA	75.45 ± 0.29 (30)	79.27 ± 1.88 (30)	82.59 ± 1.37 (27)	82.81 ± 0.43 (30)	83.54 ± 0.89 (27)	84.24 ± 0.77 (30)	84.26 ± 0.89 (30)	83.62 ± 0.72 (27)	$\mathbf{84.92 \pm 0.59}$ (30, 0.1)
		Kappa	81.21 ± 0.13 (30)	83.28 ± 0.44 (30)	85.82 ± 0.39 (27)	85.76 ± 0.20 (30)	86.13 ± 0.21 (27)	85.94 ± 0.21 (30)	86.10 ± 0.14 (30)	86.07 ± 0.21 (27)	$\mathbf{87.12 \pm 0.09}$ (30, 0.1)
	RF	OA	86.83 ± 0.14 (30)	86.76 ± 0.28 (30)	88.94 ± 0.25 (27)	88.04 ± 0.22 (30)	87.73 ± 0.32 (27)	88.30 ± 0.21 (30)	87.97 ± 0.22 (30)	89.10 ± 0.19 (30)	$\mathbf{89.38 \pm 0.11}$ (24, 10)
		AA	81.49 ± 0.36 (27)	82.83 ± 0.43 (30)	85.68 ± 0.36 (27)	84.52 ± 0.27 (30)	84.18 ± 0.36 (27)	84.88 ± 0.62 (30)	84.29 ± 0.54 (30)	85.51 ± 0.57 (24)	$\mathbf{85.75 \pm 0.36}$ (24, 10)
		Kappa	83.22 ± 0.16 (30)	83.18 ± 0.36 (30)	85.84 ± 0.33 (27)	84.73 ± 0.27 (30)	84.35 ± 0.40 (27)	85.05 ± 0.27 (30)	84.63 ± 0.28 (30)	86.03 ± 0.24 (30)	$\mathbf{86.39 \pm 0.15}$ (24, 10)

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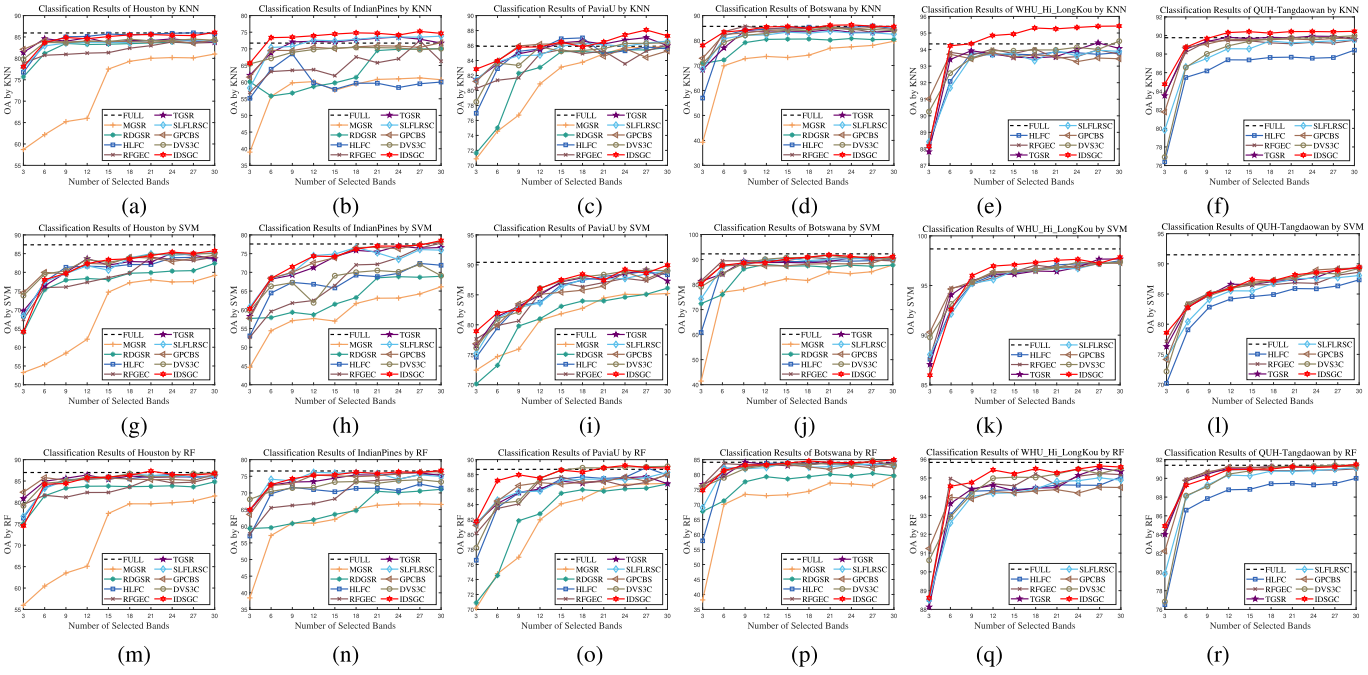


Fig. 2. OA curves on all HSI datasets by KNN (the first row), SVM (the second row), and RF (the third row), respectively. (a) Houston. (b) IndianPines. (c) PaviaU. (d) Botswana. (e) WHU-Hi-LongKou. (f) QUH-Tangdaowan. (g) Houston. (h) IndianPines. (i) PaviaU. (j) Botswana. (k) WHU-Hi-LongKou. (l) QUH-Tangdaowan. (m) Houston. (n) IndianPines. (o) PaviaU. (p) Botswana. (q) WHU-Hi-LongKou. (r) QUH-Tangdaowan.

3) The consistent and comprehensive performance improvement achieved by IDSGC across HSI datasets and classifiers demonstrates its effectiveness and advancement over existing band selection methods. IDSGC achieves the highest accuracy on all HSI datasets using both KNN and RF classifiers. Compared to the second best method, IDSGC improves the OA by 1.69% and 1.02% on the IndianPines and PaviaU with KNN, respectively. When using SVM, the OA enhancements on IndianPines and PaivaU are 0.43% and 0.80%, while 0.95% and 0.28% with RF. Compared to clustering-based postprocessing methods, i.e., TGSR, SLFLRSC, and DVS3C, our IDSGC achieves significant performance gains. Taking IndianPines, WHU-Hi-LongKou, and QUH-Tangdaowan as examples, the average improvements of OA are 3.22%, 0.38%, 0.56% with SVM, 1.90%, 0.29%, and 0.20% with RF. In particular, when performing classification with KNN, the results of our IDSGC on these three datasets are better than the clustering-based postprocessing methods in almost all land-covers, and the AA are enhanced by an average of 2.54%, 2.98%, and 1.81% on IndianPines, WHU-Hi-LongKou, and QUH-Tangdaowan.

What's more, we conduct a t-test decision for the null hypothesis that the results of our IDSGC and the best competitors are Undifferentiated. With the test decisions and p-values shown in Table VI, it can be seen that the proposed method surpasses most of the best competitors, which demonstrates the performance improvement of statistical significance. Overall, the consistent and comprehensive performance improvement achieved by IDSGC across all datasets and classifiers demonstrates its effectiveness and advancement over existing band selection methods.

Since there is no prior information to determine the optimal number of bands, we illustrate the OA curves with varying band numbers in Fig. 2, with our method represented by the red curve. In addition, the classification accuracy using all bands (denoted as FULL) is indicated by a short black horizontal line in the figures. From these curves, it is evident that IDSGC outperforms the competitors across most band number configurations. When using the KNN classifier, IDSGC obtains comparable or better classification performance than that of FULL, with more than nine bands selected. Particularly, on IndianPines, IDSGC surpasses the FULL with just six bands. For SVM classification, the proposed IDSGC also demonstrates superior performance compared to the other methods, yielding results that are closest to those obtained with FULL. While performing classification with RF, the performance of IDSGC is basically on par with FULL and better than most of the competitors.

The experimental results and analyses based on Tables III–V and Fig. 2 validate the superiority of our IDSGC method. These results demonstrate that our method effectively constructs a well-structured affinity graph and selects discriminative bands from the original band set.

C. Runtime Experiments

The computational time of our method, along with the comparison methods, is summarized in Table VII, with 30 bands selected. Across all four datasets, our method demonstrates satisfactory speed, particularly on IndianPines and PaviaU. For datasets with large spatial dimensions (Houston and Botswana) or a large number of land-cover pixels (WHU-Hi-LongKou

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TABLE IV

PER CLASS AA, OA, AA, AND KAPPA ON BOTSWANA AND WHU-HI-LONGKOU DATASETS (MEAN ± STD). FOR THE COMPARED METHODS, THE RELATED BAND NUMBERS ARE RECORDED IN PARENTHESES. FOR OUR METHOD, THE RELATED BAND NUMBER AND PARAMETER γ VALUE ARE RECORDED IN PARENTHESES. THE MAXIMUM VALUES ARE MARKED IN BOLD. "-" MEANS OUT-OF-MEMORY ERROR

Data	Classifier	Metric	MGSR(2022)	RDGSR(2022)	HLFC(2023)	RFGEC(2024)	TGSR(2024)	SLFLRSC(2024)	GPCBS(2025)	DVS3C(2025)	Our
Botswana	KNN	1	**100.00 ± 0.00**	99.64 ± 0.68	99.60 ± 0.71	98.79 ± 1.54	99.92 ± 0.18	99.40 ± 0.62	99.46 ± 0.80	99.18 ± 0.77	99.92 ± 0.18
		2	58.46 ± 11.92	71.78 ± 4.62	82.65 ± 5.62	**91.01 ± 7.26**	85.71 ± 4.11	80.42 ± 12.63	87.73 ± 4.75	87.47 ± 7.79	90.33 ± 3.68
		3	89.29 ± 1.81	92.52 ± 3.46	96.03 ± 2.05	95.24 ± 1.91	93.63 ± 2.14	95.47 ± 1.27	94.29 ± 2.52	93.36 ± 2.14	**97.35 ± 1.93**
		4	86.91 ± 2.35	82.81 ± 2.43	94.44 ± 1.38	94.76 ± 2.56	91.65 ± 2.88	91.42 ± 3.51	91.79 ± 2.03	**95.67 ± 0.59**	90.31 ± 2.71
		5	77.86 ± 2.83	69.51 ± 5.05	**79.93 ± 3.39**	71.05 ± 5.47	75.80 ± 3.37	75.95 ± 3.79	76.42 ± 6.72	78.19 ± 2.24	74.32 ± 2.30
		6	38.35 ± 5.41	46.06 ± 6.37	56.72 ± 4.92	57.63 ± 7.68	54.49 ± 6.63	44.18 ± 3.66	54.61 ± 8.11	51.36 ± 7.29	**61.65 ± 6.96**
		7	92.14 ± 3.21	92.69 ± 1.44	92.66 ± 1.33	**95.92 ± 1.97**	92.74 ± 2.03	95.11 ± 2.27	94.66 ± 1.30	94.53 ± 1.94	95.64 ± 2.48
		8	81.53 ± 9.18	83.52 ± 2.34	**97.82 ± 1.64**	95.98 ± 3.20	95.63 ± 1.09	90.69 ± 3.69	95.01 ± 2.20	92.90 ± 4.03	94.32 ± 4.65
		9	77.10 ± 4.92	76.95 ± 3.07	78.46 ± 4.14	81.65 ± 3.05	75.97 ± 5.00	75.32 ± 8.07	78.63 ± 1.16	**82.76 ± 3.22**	80.35 ± 3.57
		10	75.36 ± 9.77	**81.38 ± 3.72**	74.74 ± 3.07	75.64 ± 5.60	75.36 ± 4.34	81.23 ± 5.31	76.38 ± 4.65	76.34 ± 3.64	81.34 ± 9.96
		11	83.56 ± 4.13	84.06 ± 5.30	**91.06 ± 3.78**	88.97 ± 5.27	87.49 ± 3.73	87.40 ± 3.90	88.33 ± 2.58	88.29 ± 2.87	88.36 ± 3.64
		12	93.74 ± 2.01	95.10 ± 1.01	94.86 ± 2.48	94.38 ± 3.40	93.62 ± 1.71	95.93 ± 2.02	**96.48 ± 0.67**	94.72 ± 2.02	92.76 ± 2.65
		13	75.37 ± 2.61	75.17 ± 5.21	81.41 ± 3.02	78.45 ± 2.07	77.11 ± 3.84	77.17 ± 4.70	78.80 ± 2.04	**82.31 ± 2.94**	79.75 ± 4.00
		14	98.37 ± 2.41	98.50 ± 0.64	**99.08 ± 0.52**	98.16 ± 0.64	98.60 ± 0.52	99.05 ± 1.27	97.71 ± 1.42	99.07 ± 0.52	96.51 ± 1.84
		OA	80.04 ± 0.79 (30)	81.22 ± 0.70 (12)	85.84 ± 0.39 (27)	85.71 ± 0.75 (9)	84.35 ± 0.52 (21)	83.71 ± 0.91 (21)	85.30 ± 0.32 (27)	85.54 ± 0.32 (21)	**86.26 ± 0.47** (24, 0.1)
		AA	80.58 ± 0.95 (30)	82.12 ± 0.74 (21)	87.10 ± 0.38 (27)	86.97 ± 0.75 (9)	85.55 ± 0.96 (21)	84.91 ± 1.04 (21)	86.45 ± 0.54 (27)	86.87 ± 0.55 (21)	**87.35 ± 0.38** (24, 0.1)
		Kappa	78.95 ± 0.81 (30)	80.16 ± 0.73 (12)	84.97 ± 0.41 (27)	84.85 ± 0.79 (9)	83.42 ± 0.54 (21)	82.75 ± 0.94 (21)	84.42 ± 0.32 (27)	84.66 ± 0.34 (21)	**85.41 ± 0.49** (24, 0.1)
	SVM	OA	87.85 ± 1.09 (30)	88.09 ± 0.90 (12)	89.95 ± 0.64 (30)	91.03 ± 0.68 (24)	90.81 ± 1.05 (27)	90.63 ± 0.82 (30)	91.17 ± 0.32 (30)	91.65 ± 0.93 (21)	**91.84 ± 0.52** (21, 0.001)
		AA	88.76 ± 1.00 (30)	89.14 ± 1.04 (12)	90.88 ± 0.67 (30)	91.93 ± 0.64 (24)	91.97 ± 0.95 (27)	91.55 ± 0.72 (30)	91.93 ± 0.45 (30)	92.45 ± 0.89 (21)	**92.46 ± 0.52** (21, 0.001)
		Kappa	87.08 ± 1.14 (30)	87.33 ± 0.94 (12)	89.28 ± 0.67 (30)	90.42 ± 0.72 (24)	90.18 ± 1.11 (27)	90.00 ± 0.86 (30)	90.56 ± 0.34 (30)	91.07 ± 0.98 (21)	**91.27 ± 0.56** (21, 0.001)
	RF	OA	79.62 ± 0.34 (30)	80.42 ± 0.95 (27)	83.85 ± 1.18 (18)	83.92 ± 0.76 (18)	84.20 ± 0.64 (30)	84.01 ± 0.86 (15)	83.93 ± 1.02 (30)	84.74 ± 0.80 (27)	**84.96 ± 0.17** (30, 100)
		AA	80.22 ± 0.38 (30)	81.30 ± 0.95 (27)	84.99 ± 0.45 (15)	84.88 ± 1.15 (24)	85.44 ± 1.06 (21)	85.12 ± 1.05 (15)	84.55 ± 1.04 (30)	85.83 ± 1.02 (27)	**85.96 ± 0.17** (30, 100)
		Kappa	78.52 ± 0.35 (30)	79.36 ± 0.98 (27)	82.91 ± 1.23 (18)	82.97 ± 0.79 (18)	83.28 ± 0.66 (30)	83.07 ± 0.90 (15)	82.99 ± 1.07 (30)	83.82 ± 0.83 (27)	**84.05 ± 0.18** (30, 100)
WHU-Hi-LongKou	KNN	1	–	–	95.99 ± 0.28	96.90 ± 0.09	97.10 ± 0.16	96.37 ± 0.19	96.19 ± 0.23	97.15 ± 0.19	**98.09 ± 0.06**
		2	–	–	72.09 ± 0.71	73.64 ± 1.21	75.08 ± 0.58	71.95 ± 0.83	72.38 ± 0.57	76.15 ± 1.00	**81.77 ± 0.74**
		3	–	–	59.05 ± 0.84	55.06 ± 1.50	60.54 ± 0.87	56.09 ± 1.62	54.65 ± 1.68	**62.60 ± 0.65**	58.25 ± 0.98
		4	–	–	95.69 ± 0.15	95.57 ± 0.13	95.96 ± 0.16	95.52 ± 0.21	95.10 ± 0.11	95.82 ± 0.15	**96.24 ± 0.12**
		5	–	–	62.23 ± 1.67	64.94 ± 1.60	65.27 ± 1.72	65.36 ± 1.49	61.90 ± 1.78	**70.07 ± 0.73**	69.69 ± 1.22
		6	–	–	99.13 ± 0.04	99.10 ± 0.11	99.29 ± 0.11	99.34 ± 0.07	99.16 ± 0.17	99.33 ± 0.05	**99.38 ± 0.13**
		7	–	–	99.96 ± 0.01	**99.97 ± 0.01**	99.96 ± 0.01	99.96 ± 0.01	99.96 ± 0.01	**99.97 ± 0.01**	**99.97 ± 0.00**
		8	–	–	84.20 ± 0.62	84.61 ± 0.45	84.26 ± 0.77	83.31 ± 0.74	84.21 ± 0.42	83.81 ± 0.99	**86.56 ± 0.33**
		9	–	–	58.73 ± 0.65	60.48 ± 1.02	63.13 ± 1.50	63.40 ± 1.30	63.44 ± 0.72	62.17 ± 0.87	**76.54 ± 0.99**
		OA	–	–	93.85 ± 0.06 (21)	94.04 ± 0.11 (18)	94.41 ± 0.11 (27)	93.96 ± 0.05 (27)	93.82 ± 0.09 (12)	94.51 ± 0.04 (30)	**95.42 ± 0.02** (30, 1)
		AA	–	–	80.79 ± 0.19 (9)	81.14 ± 0.35 (18)	82.29 ± 0.34 (27)	81.26 ± 0.17 (21)	80.78 ± 0.31 (6)	83.01 ± 0.22 (30)	**85.17 ± 0.19** (30, 1)
		Kappa	–	–	92.04 ± 0.08 (21)	92.29 ± 0.14 (18)	92.76 ± 0.15 (27)	92.18 ± 0.07 (27)	92.00 ± 0.11 (12)	92.89 ± 0.05 (30)	**94.06 ± 0.03** (30, 1)
	SVM	OA	–	–	97.49 ± 0.04 (30)	97.34 ± 0.08 (27)	97.67 ± 0.05 (30)	97.35 ± 0.07 (30)	97.30 ± 0.07 (30)	97.38 ± 0.02 (30)	**97.85 ± 0.05** (30, 10)
		AA	–	–	92.65 ± 0.22 (30)	91.96 ± 0.48 (27)	93.26 ± 0.30 (30)	92.30 ± 0.57 (30)	92.30 ± 0.41 (30)	92.15 ± 0.25 (30)	**93.71 ± 0.37** (30, 10)
		Kappa	–	–	96.73 ± 0.05 (30)	96.52 ± 0.06 (27)	96.96 ± 0.07 (30)	96.54 ± 0.09 (30)	96.48 ± 0.09 (30)	96.58 ± 0.03 (30)	**97.19 ± 0.07** (30, 10)
	RF	OA	–	–	95.06 ± 0.06 (30)	95.24 ± 0.08 (27)	95.53 ± 0.06 (27)	95.01 ± 0.08 (27)	94.51 ± 0.08 (27)	95.52 ± 0.10 (30)	**95.64 ± 0.06** (27, 10)
		AA	–	–	83.48 ± 0.22 (30)	84.35 ± 0.30 (27)	84.82 ± 0.22 (27)	83.75 ± 0.33 (27)	82.40 ± 0.25 (27)	84.75 ± 0.21 (24)	**85.19 ± 0.25** (27, 10)
		Kappa	–	–	93.59 ± 0.08 (30)	93.81 ± 0.10 (27)	94.20 ± 0.08 (27)	93.53 ± 0.10 (27)	92.89 ± 0.10 (27)	84.17 ± 0.13 (30)	**94.34 ± 0.08** (27, 10)

and QUH-Tangdaowan), most methods suffer from the issue of increased computational time, yet our IDSGC method maintains acceptable efficiency. Overall, the computational performance of our IDSGC remains competitive, even on large-scale and high-dimensional hyperspectral datasets.

D. Convergence Experiments

To validate the convergence of our optimization strategy in Algorithm 2, the convergence condition is determined by the difference value of the objective function, as given by (5). The convergence curves for four HSI datasets are plotted in Fig. 3, where the x-axis represents the iteration number and the y-axis denotes the objective function value. As seen from the figures, the objective values decrease rapidly during the first five iterations and subsequently stabilize, which confirms the convergence analysis of our model in Section II-C.

E. Parameter Sensitivity Analysis

In the proposed IDSGC method [see (5)], the only hyperparameter to tune is the balanced parameter γ. We define the search grid for γ as $\{10^{-3}, 10^{-2}, 10^{-1}, \ldots, 10^{3}\}$, and the results of sensitivity test are recorded in Fig. 4. On all benchmark HSI datasets, we plot the OA curves for all γ values across varying numbers of selected bands. When the number of bands is fixed, the performance differences caused by different γ values are minimal in Houston and Botswana. As the number of

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TABLE V

PER CLASS AA, OA, AA, AND KAPPA ON QUH-TANGDAOWAN DATASET (MEAN ± STD). FOR THE COMPARED METHODS, THE RELATED BAND NUMBERS ARE RECORDED IN PARENTHESES. FOR OUR METHOD, THE RELATED BAND NUMBER AND PARAMETER γ VALUE ARE RECORDED IN PARENTHESES. THE MAXIMUM VALUES ARE MARKED IN BOLD. "-" MEANS OUT-OF-MEMORY ERROR

Data	Classifier	Metric	MGSR(2022)	RDGSR(2022)	HLFC(2023)	RFGEC(2024)	TGSR(2024)	SLFLRSC(2024)	GPCBS(2025)	DVS3C(2025)	Our
QUH-Tangdaowan	KNN	1	–	–	98.30 ± 0.19	99.10 ± 0.07	$\mathbf{99.38 \pm 0.09}$	99.23 ± 0.06	99.15 ± 0.22	99.26 ± 0.13	99.36 ± 0.14
		2	–	–	93.90 ± 0.13	94.97 ± 0.23	95.47 ± 0.06	95.06 ± 0.15	95.16 ± 0.18	95.29 ± 0.22	$\mathbf{95.94 \pm 0.09}$
		3	–	–	86.50 ± 0.25	89.73 ± 0.39	90.70 ± 0.23	89.83 ± 0.32	90.71 ± 0.33	90.17 ± 0.27	$\mathbf{90.77 \pm 0.37}$
		4	–	–	97.76 ± 0.12	98.18 ± 0.07	98.35 ± 0.13	98.13 ± 0.19	98.18 ± 0.07	98.19 ± 0.18	$\mathbf{98.42 \pm 0.14}$
		5	–	–	80.94 ± 1.83	85.38 ± 1.47	87.09 ± 1.22	85.02 ± 1.65	$\mathbf{87.52 \pm 1.35}$	87.41 ± 0.62	85.88 ± 1.15
		6	–	–	80.29 ± 0.38	82.61 ± 0.42	83.41 ± 0.26	82.97 ± 0.45	83.31 ± 0.39	83.01 ± 0.45	$\mathbf{83.65 \pm 0.55}$
		7	–	–	66.51 ± 1.01	67.47 ± 0.50	69.01 ± 0.32	67.83 ± 0.52	69.68 ± 0.25	69.63 ± 0.79	$\mathbf{72.50 \pm 0.71}$
		8	–	–	98.88 ± 0.12	99.03 ± 0.14	99.20 ± 0.09	99.08 ± 0.11	99.10 ± 0.06	99.17 ± 0.06	$\mathbf{99.24 \pm 0.08}$
		9	–	–	90.85 ± 0.50	92.41 ± 0.20	92.63 ± 0.25	92.05 ± 0.35	92.47 ± 0.21	92.53 ± 0.27	$\mathbf{93.19 \pm 0.14}$
		10	–	–	60.57 ± 1.82	62.65 ± 0.79	65.96 ± 1.30	65.78 ± 1.92	64.34 ± 2.44	65.66 ± 2.23	$\mathbf{70.07 \pm 2.03}$
		11	–	–	39.33 ± 0.45	41.18 ± 0.73	43.21 ± 0.96	41.23 ± 0.50	42.17 ± 0.23	41.86 ± 0.48	$\mathbf{46.35 \pm 0.27}$
		12	–	–	38.31 ± 2.43	41.57 ± 3.07	44.86 ± 1.44	40.92 ± 2.77	44.71 ± 2.43	42.43 ± 3.02	$\mathbf{49.54 \pm 2.42}$
		13	–	–	93.39 ± 0.45	94.28 ± 1.45	94.41 ± 1.22	94.31 ± 1.14	94.24 ± 0.59	$\mathbf{94.56 \pm 0.16}$	94.47 ± 1.57
		14	–	–	60.30 ± 2.61	61.08 ± 1.91	65.19 ± 2.24	62.11 ± 1.76	61.40 ± 1.86	64.76 ± 2.08	$\mathbf{65.66 \pm 1.30}$
		15	–	–	63.97 ± 0.93	69.99 ± 0.59	70.65 ± 0.53	67.65 ± 0.85	69.68 ± 0.43	70.47 ± 1.12	$\mathbf{73.57 \pm 0.55}$
		16	–	–	90.10 ± 0.22	90.33 ± 0.08	90.45 ± 0.08	90.15 ± 0.12	90.47 ± 0.13	90.33 ± 0.22	$\mathbf{90.54 \pm 0.20}$
		17	–	–	47.51 ± 0.59	53.07 ± 0.85	54.11 ± 0.55	52.15 ± 0.60	51.91 ± 0.20	53.55 ± 0.51	$\mathbf{58.48 \pm 0.49}$
		18	–	–	99.56 ± 0.04	99.57 ± 0.03	99.56 ± 0.05	99.58 ± 0.05	99.61 ± 0.04	99.64 ± 0.05	$\mathbf{99.70 \pm 0.04}$
		OA	–	–	88.42 ± 0.05 (30)	89.49 ± 0.03 (30)	89.93 ± 0.04 (24)	89.52 ± 0.04 (30)	89.75 ± 0.03 (30)	89.98 ± 0.06 (30)	$\mathbf{90.45 \pm 0.04}$ (30, 1)
		AA	–	–	77.05 ± 0.42 (30)	79.03 ± 0.20 (30)	80.20 ± 0.16 (24)	79.06 ± 0.21 (27)	79.66 ± 0.14 (30)	79.88 ± 0.10 (21)	$\mathbf{81.52 \pm 0.23}$ (30, 1)
		Kappa	–	–	87.10 ± 0.05 (30)	88.28 ± 0.04 (30)	88.76 ± 0.04 (24)	88.31 ± 0.04 (30)	88.57 ± 0.03 (30)	88.81 ± 0.07 (30)	$\mathbf{89.33 \pm 0.05}$ (30, 1)
	SVM	OA	–	–	87.35 ± 0.07 (30)	88.61 ± 0.06 (30)	89.30 ± 0.04 (30)	88.07 ± 0.38 (30)	89.31 ± 0.07 (30)	89.25 ± 0.05 (30)	$\mathbf{89.44 \pm 0.05}$ (30, 1)
		AA	–	–	77.30 ± 0.46 (30)	78.93 ± 0.36 (30)	80.78 ± 0.12 (30)	77.29 ± 1.03 (30)	80.59 ± 0.16 (30)	79.74 ± 0.73 (30)	$\mathbf{80.83 \pm 0.35}$ (30, 1)
		Kappa	–	–	85.84 ± 0.08 (30)	87.23 ± 0.07 (30)	87.99 ± 0.04 (30)	86.61 ± 0.42 (30)	88.00 ± 0.08 (30)	87.93 ± 0.05 (30)	$\mathbf{88.15 \pm 0.06}$ (30, 1)
	RF	OA	–	–	90.01 ± 0.08 (30)	91.01 ± 0.05 (30)	91.35 ± 0.08 (27)	91.13 ± 0.06 (30)	91.30 ± 0.08 (21)	91.41 ± 0.05 (27)	$\mathbf{91.50 \pm 0.07}$ (30, 1)
		AA	–	–	78.42 ± 0.52 (30)	80.54 ± 0.32 (30)	81.15 ± 0.57 (27)	81.09 ± 0.14 (30)	81.27 ± 0.25 (27)	81.48 ± 0.37 (27)	$\mathbf{81.73 \pm 0.44}$ (30, 1)
		Kappa	–	–	88.80 ± 0.09 (30)	89.92 ± 0.06 (30)	90.29 ± 0.09 (27)	90.05 ± 0.06 (30)	90.23 ± 0.09 (21)	90.36 ± 0.05 (27)	$\mathbf{90.46 \pm 0.07}$ (30, 1)

TABLE VI

TEST DECISIONS AND P-VALUES FROM T-TEST COMPARING THE PROPOSED METHOD AGAINST THE BEST COMPETITORS ON ALL DATASETS. THE LEVEL OF SIGNIFICANCE IS SET TO 0.05. THE NULL HYPOTHESIS INDICATES THAT THERE IS NO DIFFERENCE IN THE RESULTS, AND THE RESULT H IS 1 IF THE TEST REJECTS THE NULL HYPOTHESIS

Classifier	Houston	IndianPines	PaviaU	Botswana	WHU-Hi-LongKou	QUH-Tangdaowan
KNN	HLFC h=0,p=0.8175	TGSR **h=1,p=0.0002**	TGSR **h=1,p=0.0064**	HLFC **h=1,p=0.0139**	DVS3C **h=1,p=2E-7**	DVS3C **h=1,p=4E-7**
SVM	SLFLRSC h=0,p=0.5178	GPCBS **h=1,p=0.0297**	TGSR **h=1,p=1E-5**	DVS3C h=0,p=0.2180	TGSR **h=1,p=0.0008**	GPCBS **h=1,p=0.0073**
RF	DVS3C **h=1,p=0.0039**	GPCBS **h=1,p=0.0126**	DVS3C **h=1,p=0.0214**	DVS3C h=0,p=0.7766	TGSR **h=1,p=0.0384**	DVS3C h=0,p=0.3467

selected bands increases, performance generally improves and then stabilizes. When selecting 30 bands, γ values of $10^0, 10^1$ tend to yield prominent performance in most experiments.

F. Ablation Experiments

IDSGC consists of two key components, which are integrated through the ILM of a doubly stochastic graph. To comprehensively verify the effect of IDSGC, we conduct ablation studies focusing on the following four components.

1) Robustness to Different Initial Graphs:

The learning of a doubly stochastic matrix optimizes the graph structure, thereby enhancing the robustness to different initial graphs. To assess this, we compare the band selection results based on different initial graphs W and the corresponding doubly stochastic graphs $Z^T\Lambda^{-1}Z$, as shown in Table VIII. On all the HSI datasets, the band subsets selected using $Z^T\Lambda^{-1}Z$ outperform those selected with W, indicating that the band structure described by $Z^T\Lambda^{-1}Z$ is clearer and more accurate. Furthermore, the classification results based on $Z^T\Lambda^{-1}Z$ remain consistent across different initialization methods. Even when the initial graphs are noisy, the learned $Z^T\Lambda^{-1}Z$ still yields stable classification results, for example, using Graph2 on PaviaU and Botswana datasets. This demonstrates that the learned doubly stochastic graph is robust to the initial graph and can reliably uncover the accurate affinity structure.

2) Contributions of Joint Learning Strategy:

IDSGC consists of two components, LBGM and subspace learning, which are integrated through implicit doubly stochastic graph learning. The joint learning strategy of Z and $Z^T\Lambda^{-1}Z$ is central for the proposed IDSGC method, enabling

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TABLE VII

RUNNING TIME OF DIFFERENT METHODS ON HSI DATASETS. THE BAND NUMBER IS 30. "-" MEANS OUT-OF-MEMORY ERROR (MEASURED BY SECONDS)

Method	Houston	IndianPines	PaviaU
MGSR	23.5036	6.8913	138.2801
RDGSR	31.1029	10.5016	203.0679
HLFC	24.6600	20.6753	7.7027
RFGEC	89.2040	2.2247	11.2986
TGSR	149.6129	17.9315	19.2710
SLFLRSC	12.5305	2.0084	2.6265
GPCBS	1030.8223	53.1604	200.8789
DVS3C	60.0168	4.5185	12.9528
Our	34.3124	2.0185	8.9594

Method	Botswana	WHU-Hi-LongKou	QUH-Tangdaowan
MGSR	3.1316	-	-
RDGSR	3.3111	-	-
HLFC	19.7643	58.3067	95.3753
RFGEC	24.8394	57.0633	165.2306
TGSR	55.6558	124.3252	443.6552
SLFLRSC	5.7913	15.4486	38.9352
GPCBS	624.1757	2843.5731	5782.5504
DVS3C	34.0689	49.4917	254.7913
Our	19.2514	12.1880	61.3154

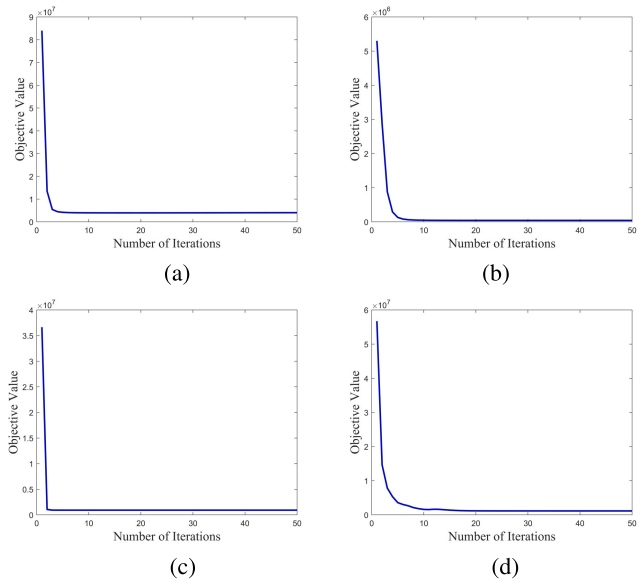


Fig. 3. Convergence curves of IDSGC on four HSI datasets. (a) Houston. (b) IndianPines. (c) PaviaU. (d) Botswana.

simultaneously exploration of both the latent representation and affinity structure information within HSI data. To validate the impact of each component, we conduct an ablation study as shown in Table IX. The results reveal that the absence of any component leads to a decrease in classification results. This limitation arises because the ablation model fails to preserve both data representations and graph structure simultaneously.

3) Advantages of the Graph Normalization in Learning:

We validate the advantage of the implicit doubly stochastic graph learning strategy by comparing it to other postprocessing methods applied to the coefficient matrix Z. The results are shown in Table X. The commonly used postprocessing

TABLE VIII

ABLATION STUDY OF IMPLICIT DOUBLY STOCHASTIC GRAPH LEARNING ON INITIAL GRAPHS. THE GRAPH INITIALIZATION MANNERS ARE LISTED IN SECTION II-C

DataSet	Initial Graph	KNN		SVM	
		W	$Z^T\Lambda^{-1}Z$	W	$Z^T\Lambda^{-1}Z$
Houston	Graph1	84.73	**85.00**	84.04	**86.05**
	Graph2	80.19	**84.14**	79.54	**84.46**
	Graph3	85.00	**85.54**	84.09	**85.84**
	Graph4	83.60	**85.20**	81.16	**84.32**
IndianPines	Graph1	68.73	**74.10**	76.03	**76.97**
	Graph2	68.58	**71.87**	76.55	**77.12**
	Graph3	68.94	**72.31**	75.16	**77.15**
	Graph4	69.52	**72.74**	74.70	**75.07**
PaviaU	Graph1	83.56	**88.00**	88.09	**89.92**
	Graph2	76.57	**84.15**	78.90	**89.60**
	Graph3	86.01	**87.59**	88.28	**90.12**
	Graph4	84.25	**84.62**	87.80	**89.63**
Botswana	Graph1	84.36	**85.65**	90.42	**91.21**
	Graph2	60.18	**86.23**	67.78	**90.89**
	Graph3	84.87	**85.39**	89.68	**90.25**
	Graph4	81.36	**85.17**	80.30	**89.91**

TABLE IX

ABLATION STUDY OF IDSGC WITH DIFFERENT COMPONENTS ON HSI DATASETS

Method	KNN			
	Houston	IndianPines	PaviaU	Botswana
w.o. Subspace Learning	85.65	68.23	82.94	84.82
w.o. LBGM	85.78	67.01	83.74	85.09
IDSGC	**86.56**	**70.33**	**85.60**	**85.99**

Method	SVM			
	Houston	IndianPines	PaviaU	Botswana
w.o. Subspace Learning	84.16	76.39	87.19	89.92
w.o. LBGM	84.92	76.51	88.99	91.00
IDSGC	**84.94**	**77.08**	**89.98**	**91.26**

TABLE X

ABLATION STUDY OF IDSGC WITH DIFFERENT POSTMANNERS

Post Manner	KNN			
	Houston	IndianPines	PaviaU	Botswana
ℓ_2 Norm	79.72	62.36	82.77	79.88
SymGraph	84.81	68.33	83.20	84.90
$Z^T\Lambda^{-1}Z$	**85.65**	**73.58**	**85.68**	**86.42**

Post Manner	SVM			
	Houston	IndianPines	PaviaU	Botswana
ℓ_2 Norm	74.90	54.57	76.57	87.26
SymGraph	85.10	77.14	88.17	90.74
$Z^T\Lambda^{-1}Z$	**86.73**	**77.65**	**89.33**	**91.71**

manners compared in this ablation experiment are ℓ_2-norm calculation (ℓ_2-Norm) and symmetric graph (SymGraph) construction, as listed in Table II.

Among these three manners, ℓ_2-Norm calculates the band importance but cannot mitigate the influence of band redundancy, leading to generally lower accuracy. SymGraph

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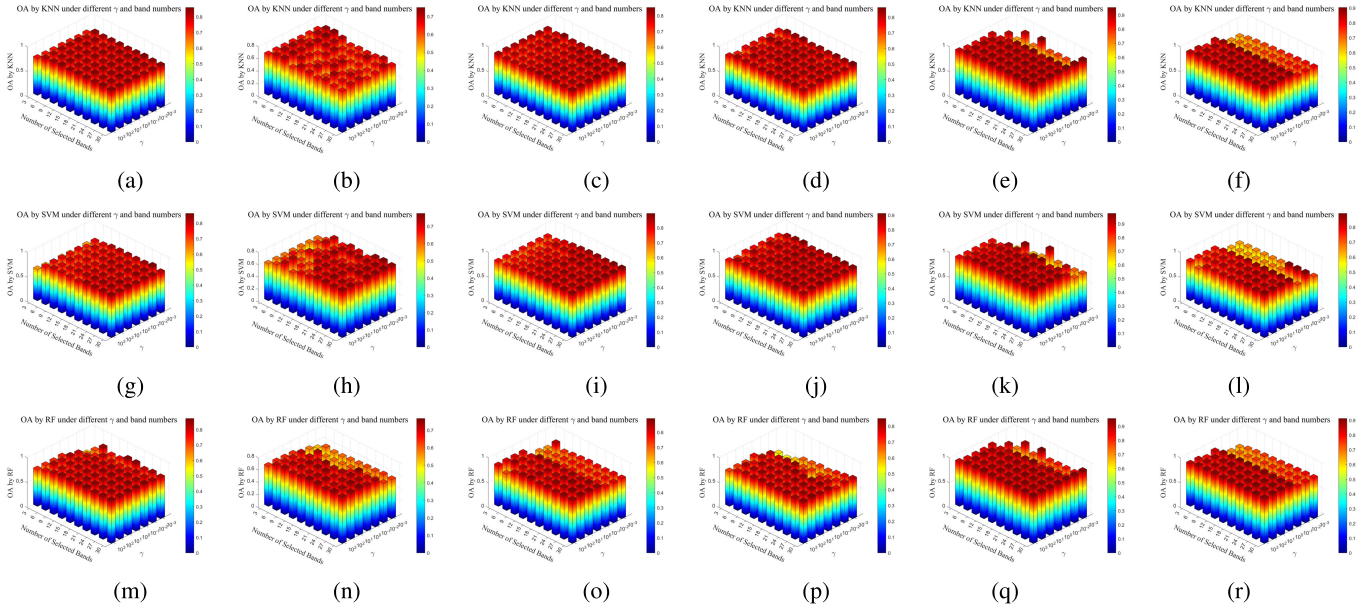


Fig. 4. Parameter sensitivity analysis curves in terms of OA on all HSI datasets by KNN (the first row), SVM (the second row), and RF (the third row). (a) Houston. (b) IndianPines. (c) PaviaU. (d) Botswana. (e) WHU-Hi-LongKou. (f) QUH-Tangdaowan. (g) Houston. (h) IndianPines. (i) PaviaU. (j) Botswana. (k) WHU-Hi-LongKou. (l) QUH-Tangdaowan. (m) Houston. (n) IndianPines. (o) PaviaU. (p) Botswana. (q) WHU-Hi-LongKou. (r) QUH-Tangdaowan.

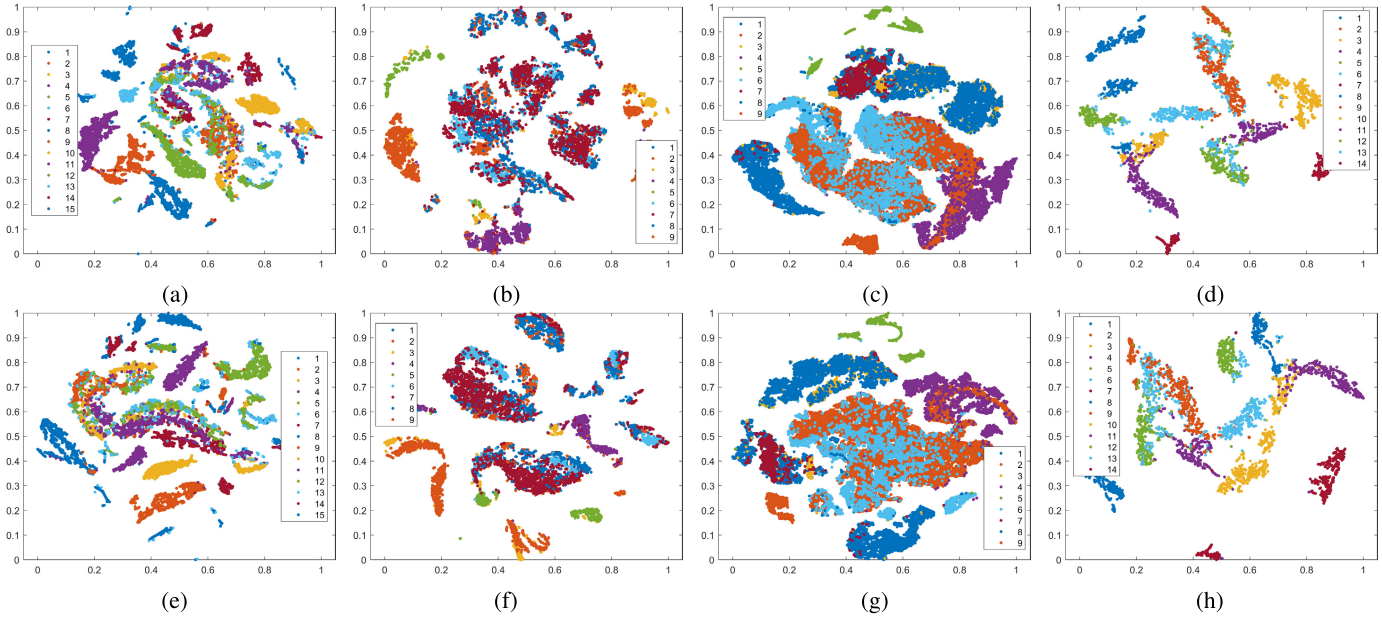


Fig. 5. t-SNE visualization of the original band distributions and the selected band distributions. (a) Original Houston. (b) Original IndianPines. (c) Original PaviaU. (d) Original Botswana. (e) Selected Houston. (f) Selected IndianPines. (g) Selected PaviaU. (h) Selected Botswana.

construction is a separate process that results in a suboptimal graph structure, performing worse than the doubly stochastic graph. In contrast, the $Z^T\Lambda^{-1}Z$ integrates graph normalization into the optimization process, ensuring symmetric and probability constraints. Thus, the improved normalization manner yields the highest OA, demonstrating the effectiveness of the ILM of a doubly stochastic graph.

4) Advantages of the Doubly Stochastic Normalization:

Here, we test the importance of doubly stochastic normalization in a manner with the results reported in Table XI. We replace the term and constraint corresponding to (3) in (5) with those in (2), to construct the ablation model (noted as NonDS). The doubly stochastic constraints in our model can effectively reconstruct band affinities in a low-dimensional space, preserving essential connections while mitigating noisy links. The absence of doubly stochastic constraints leads to worse classification performance.

G. t-SNE Visualization

To visualize the ability of the IDSGC method, we project the selected bands into 2-D subspaces obtained by t-SNE. We compare the Houston, IndianPines, PaviaU, and Botswana

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TABLE XI

ABLATION STUDY OF THE DOUBLY STOCHASTIC NORMALIZATION. NONDS IS THE ABLATION MODEL WITHOUT DOUBLY STOCHASTIC CONSTRAINTS

DataSet	KNN		RF	
	NonDS	IDSGC	NonDS	IDSGC
Houston	84.30	**85.22**	85.21	**86.49**
IndianPines	63.08	**72.44**	74.76	**76.44**
PaviaU	85.20	**86.00**	87.61	**89.20**
Botswana	83.84	**85.87**	82.72	**84.54**
WHU-Hi-LongKou	95.01	**95.33**	94.95	**95.30**
QUH-Tangdaowan	90.11	**90.71**	90.87	**91.31**

datasets as examples; the visualization results are shown in Fig. 5. The correspondence between the numbers and the land covers could be found in Table I. From these figures, it can be seen that our method discriminates the different land covers as much as possible, and makes the feature distribution of the same land-covers more consistent. This fact intuitively demonstrates that the selected bands by the IDSGC improve the expressiveness of band features by establishing doubly stochastic connections.

IV. CONCLUSION

In this article, we propose an implicit doubly stochastic graph-based subspace clustering method to address the issues in subspace learning methods, such as the need for separate normalization and the inability to preserve data affinity. The coefficient matrix simultaneously encodes the latent representation and data similarity, prompting the design of a unified framework that integrates an LBGM module with a subspace learning module. An ILM of a doubly stochastic graph, which transforms the coefficient matrix into a doubly stochastic form, naturally bridges these two modules. Moreover, the ILM alleviates the negative impacts caused by separate graph normalization. Thus, with the doubly stochastic graph, we can obtain band clusters by employing spectral clustering on it. Experimental results, including parameter sensitivity and ablation study, validate the effectiveness of the implicit doubly stochastic graph learning strategy and the joint learning framework. Comparative experiments demonstrate that the proposed IDSGC outperforms the SOTA band selection methods.

The continuously differentiable doubly stochastic graph construction form $Z^T\Lambda^{-1}Z$ proposed in this work is not restricted to HSI processing, which could be naturally integrated into a broader graph learning framework. For instance, in deep subspace learning, many recent methods adopt a two-stage paradigm, first learning the coefficient matrix, and subsequently constructing the affinity graph $(|Z| + |Z^T|)/2$. Such a paradigm may limit the graph modeling capability of deep networks, since the learning of the coefficient matrix is not directly guided by the affinity-aware objectives. Even when $(|Z| + |Z^T|)/2$ is incorporated into the loss function, the absolute value operation introduces subgradient propagation issues, which may affect the optimization stability. In contrast, embedding the continuously differentiable form $Z^T\Lambda^{-1}Z$ into the loss term enables end-to-end and stable optimization, facilitating joint representation learning and graph construction.

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